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SAR Backscatter 101

September 2025

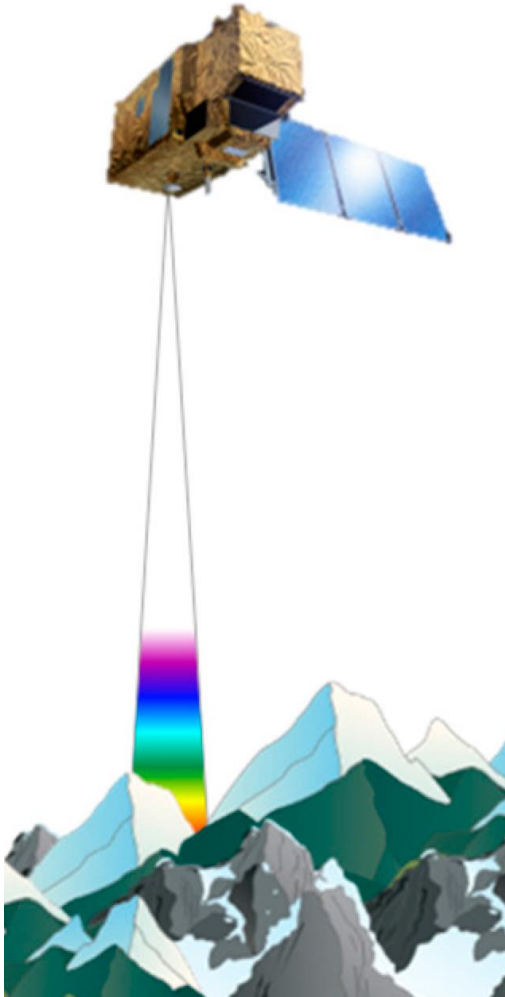
Digital Earth Antarctica & Digital Earth Australia



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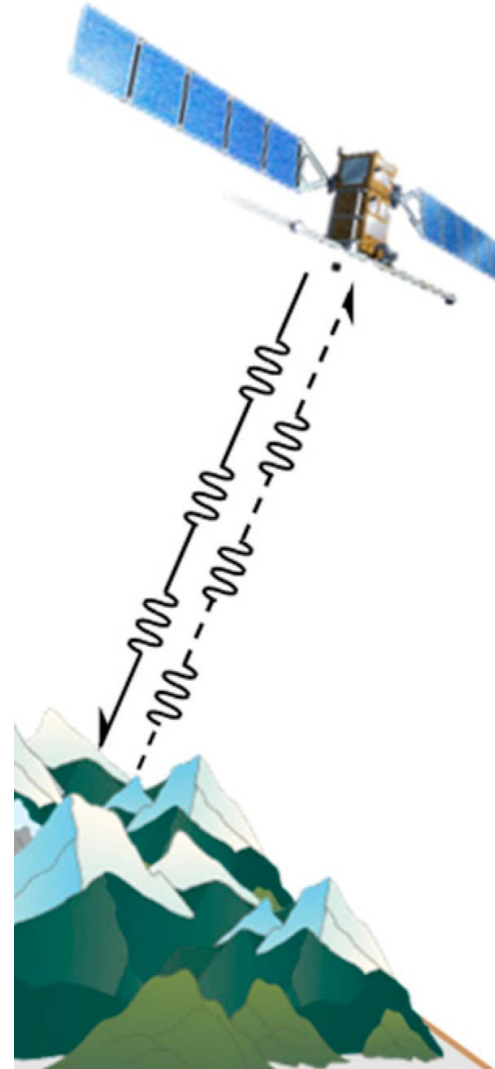
Earth sciences for Australia's future | ga.gov.au

Synthetic Aperture Radar (SAR) capability adds value



Optical Sensor

- Visible light
- Observes colour
- Passive, requires sunlight
- Observes surface only
- Obscured by clouds
- Nadir
- Simple geometry

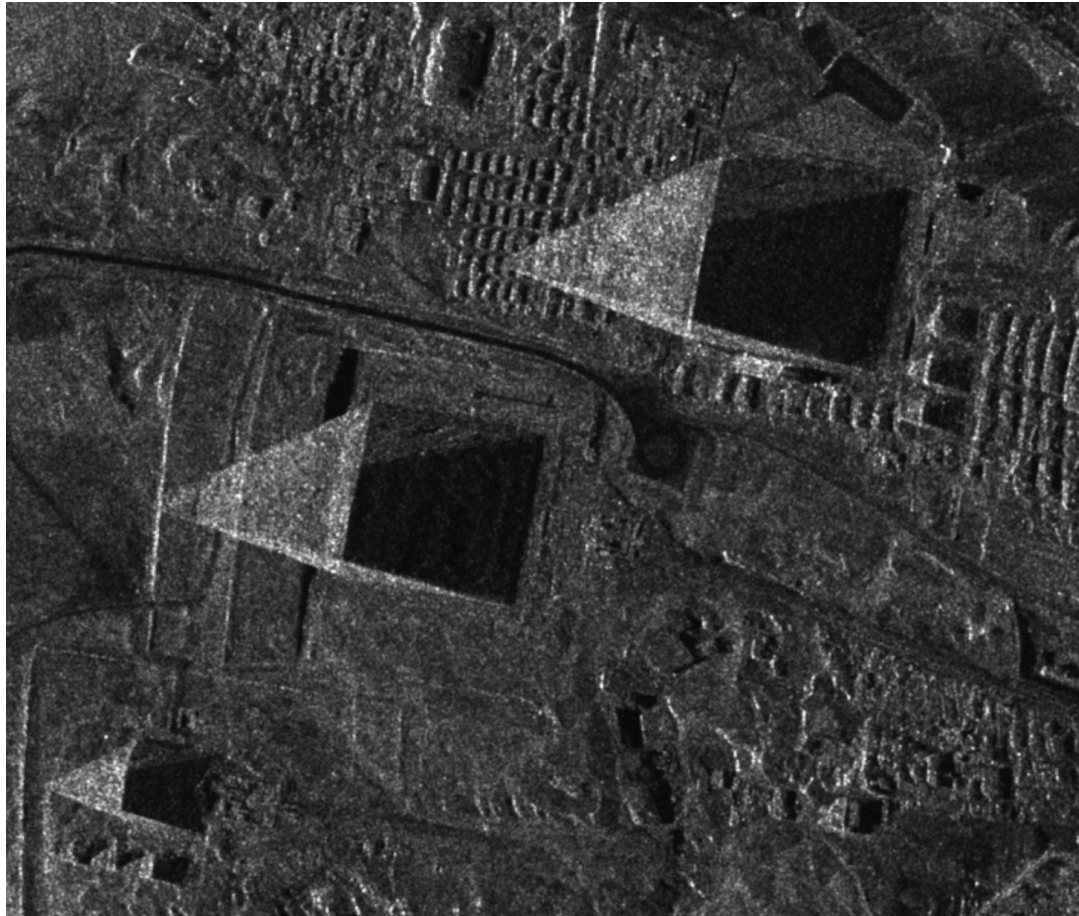


SAR Sensor

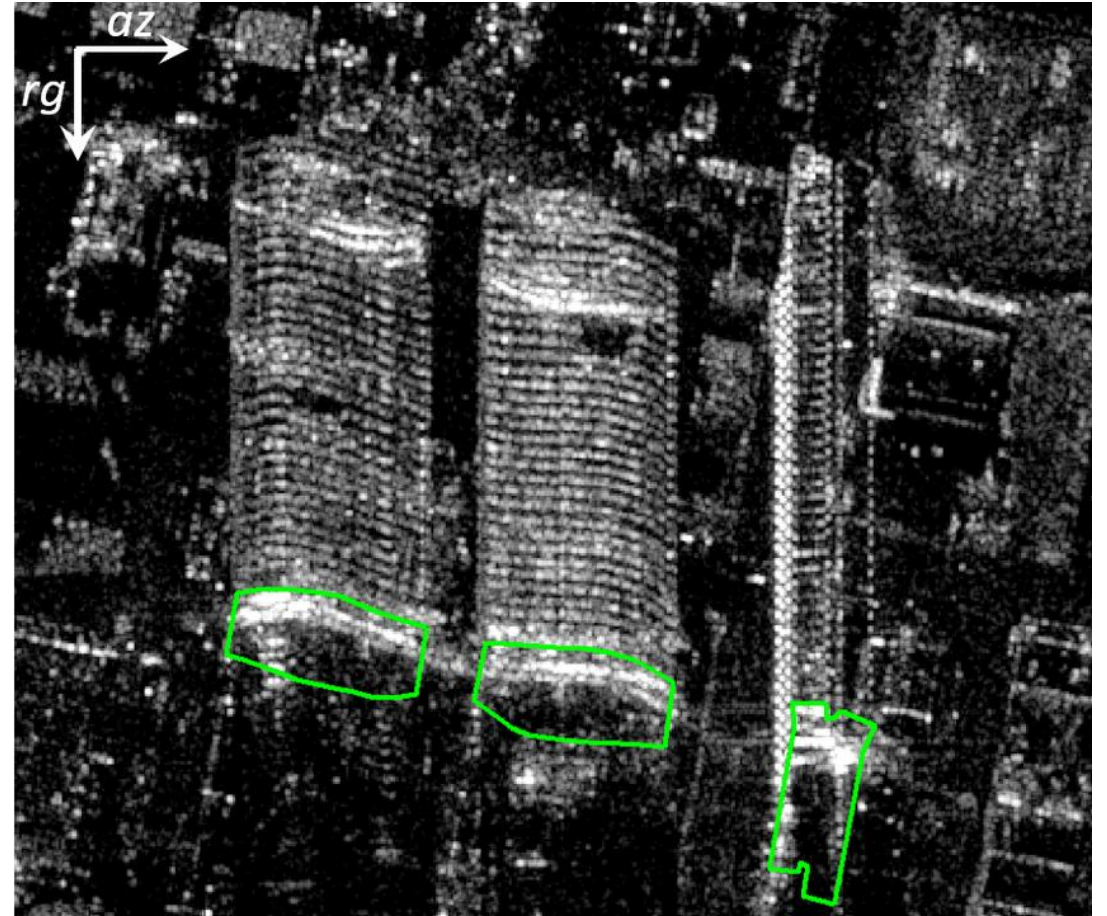
- Microwaves
- Observes shape and roughness
- Active, works day and night
- Observes dielectric properties
- Views through clouds
- Side-looking (usually)
- Complex geometry

Sometimes things look different

Gaza Pyramids

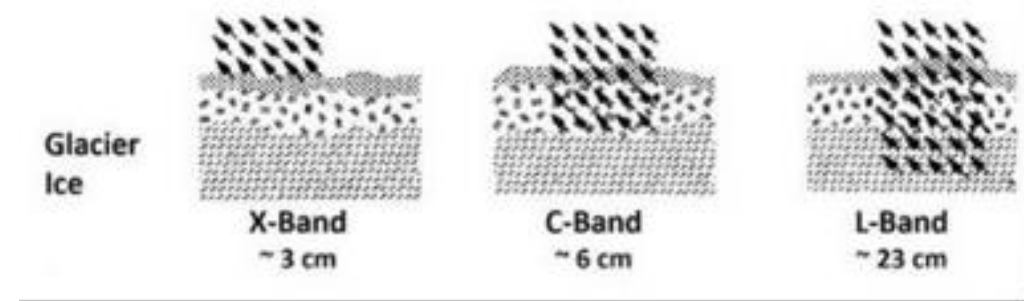
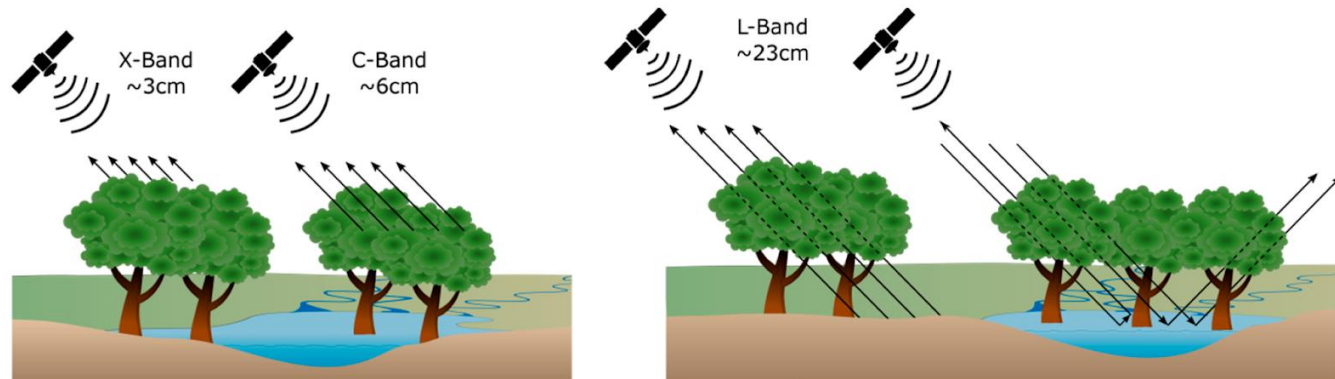
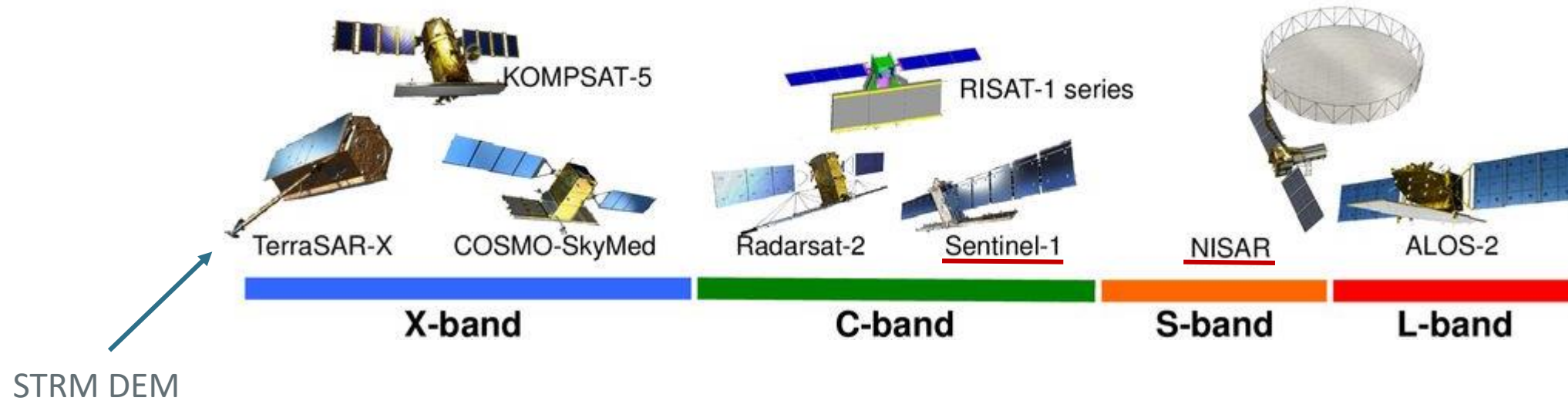


Skyscrapers



SAR Missions and Bands/Frequencies/Wavelengths

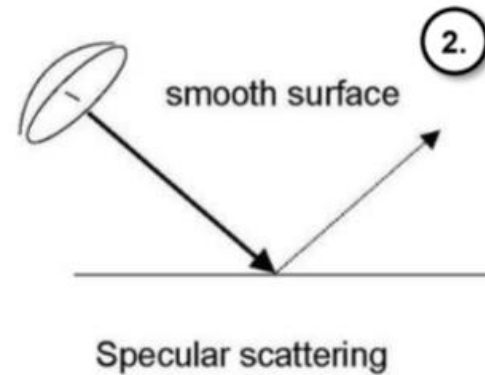
- Focus on ESA Sentinel-1 (C-Band) and NASA-ISRO NISAR (S/L-band) missions
- Bands/wavelength controls penetration of the signal



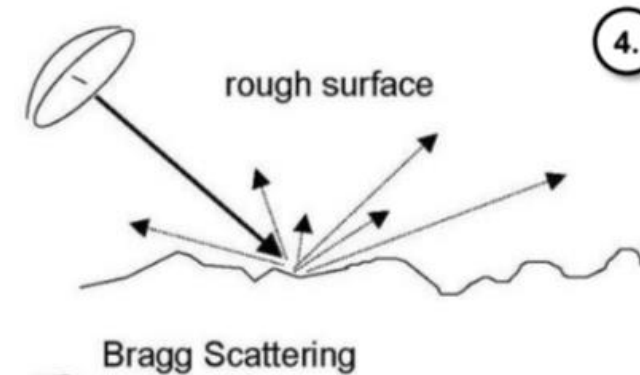
Scattering Theory

$$S_{HH} > S_{VV} \text{ close to } 0$$

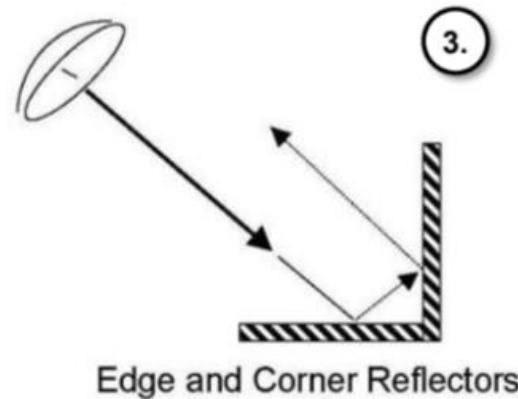
$$S_{VH} = 0$$



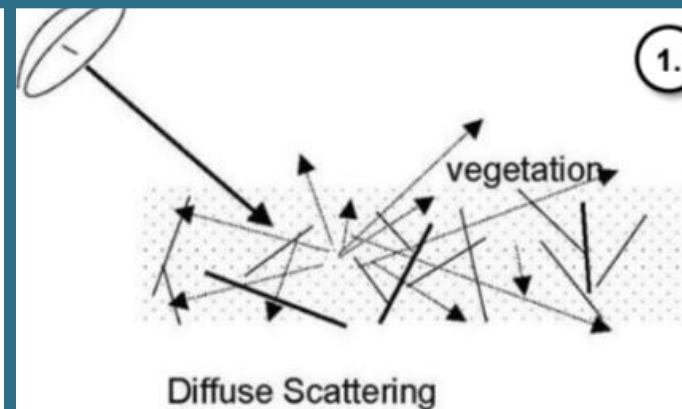
$$|S_{VV}| > |S_{HH}| > |S_{HV}| \text{ or } |S_{VH}|$$



- Scattering depends on roughness/texture
- Depends on incidence angle
- Randomly rough: Ag. fields, Low veg ...
- Periodically rough: Ocean/Waves...
- Polarisation preserving



$$|S_{HH}| > |S_{VV}| > |S_{HV}| \text{ or } |S_{VH}|$$



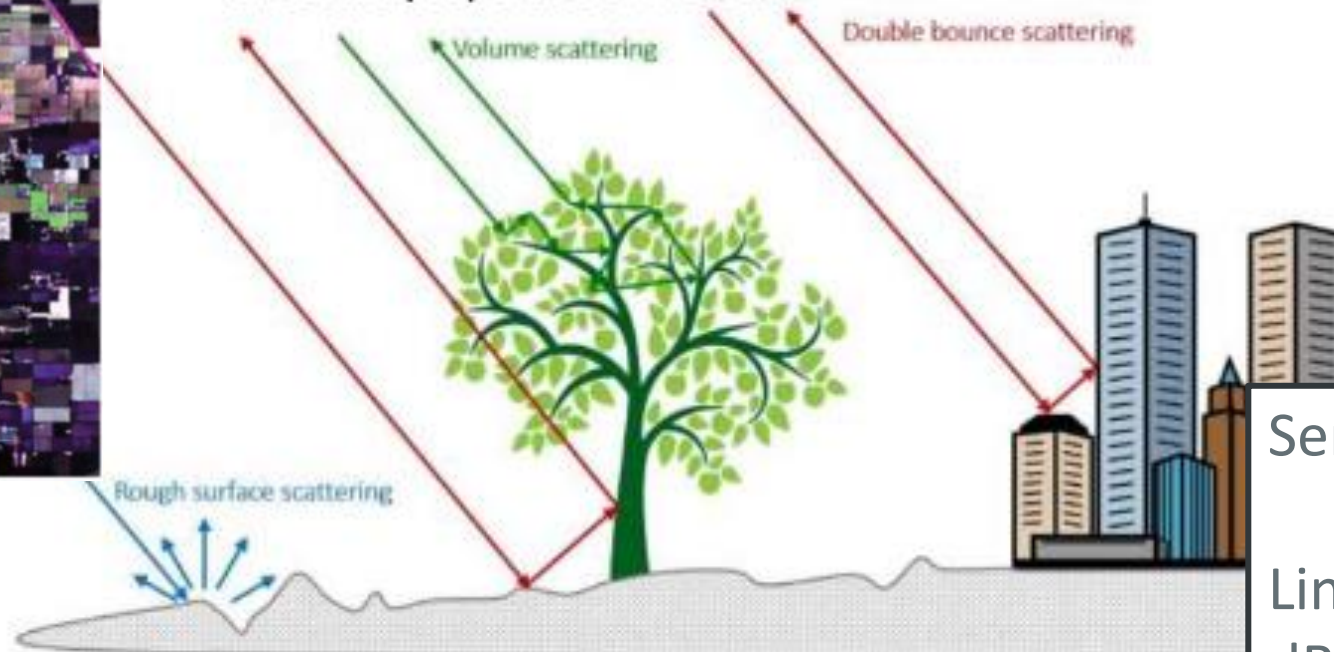
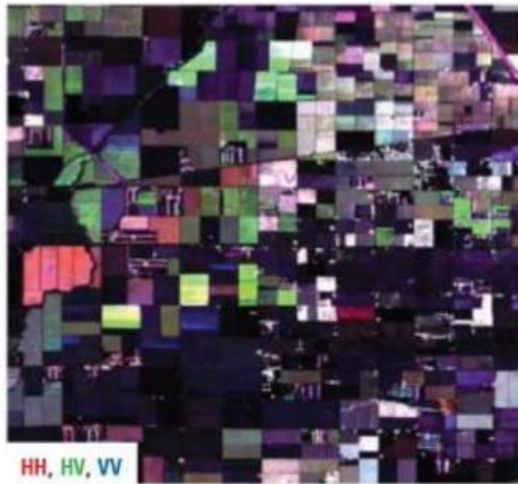
- Snow, ice, forest, dry sand agriculture fields
- High density, low size
- Return depends on:
 - density (e.g. biomass)
 - dielectric properties (i.e. moisture)
 - Thickness/depth
 - Scatterer (leaf/grain) size

main source of $|S_{HV}|$ and $|S_{VH}|$

Interpretation of Polarimetry

specular or deep volume with full attenuation : dry deep snow/sand, flat water, smooth ice

- **Scattering on (rough) surfaces**: Water, bare soils, roads – Scattering strongly dependent on surface roughness and sensor wavelength
- **Double-bounce scattering**: Buildings, tree trunks, light poles – little wavelength dependence
- **Volume Scattering**: Vegetation; dry soils with high penetration – strongly dependent on sensor wavelength and dielectric properties of medium



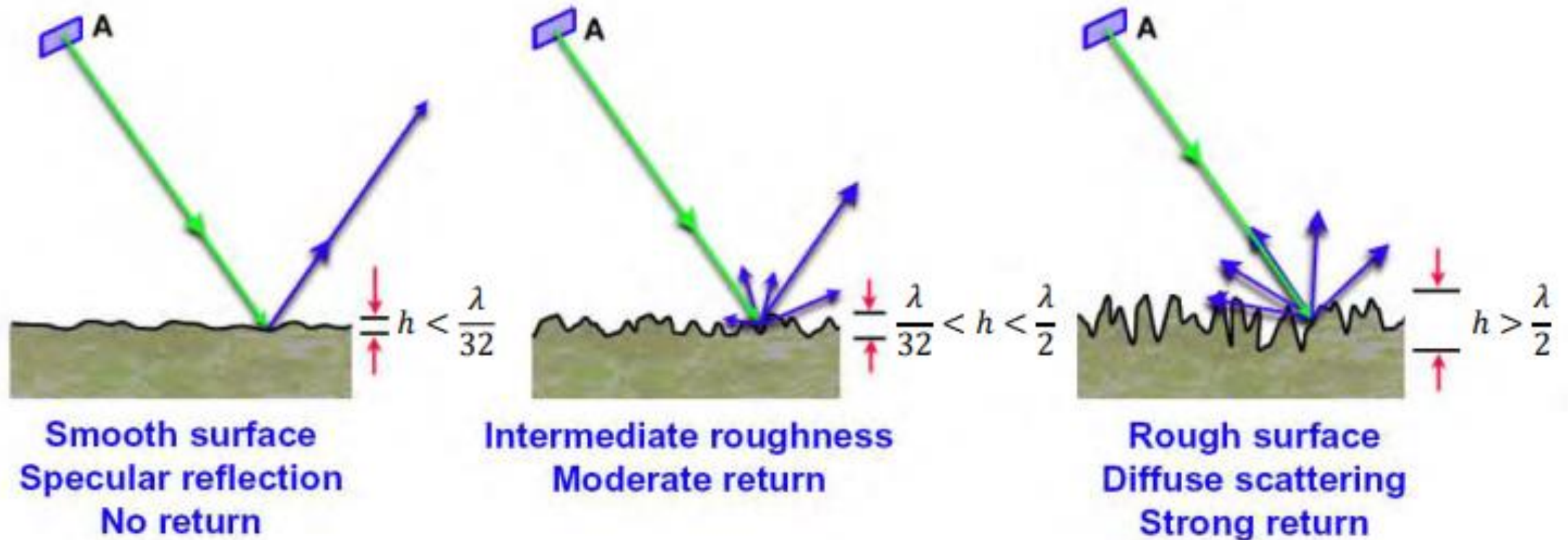
Sentinel-1 RGB decomp.

Linear: [VV, VH, VV/VH]

dB: [VV, VH, VV - VH]

A note on roughness

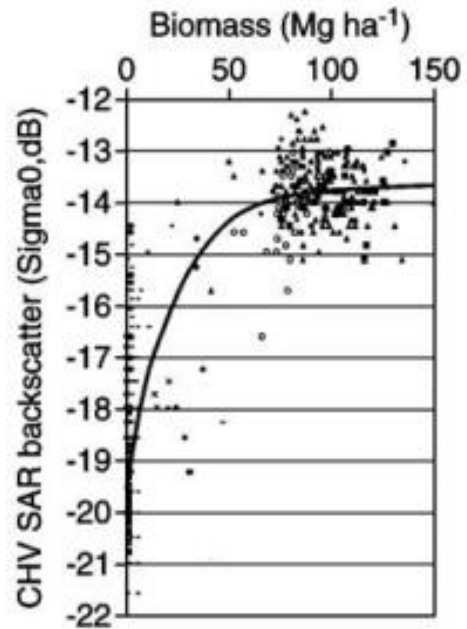
- Response relative to wavelength
- Increased roughness -> increased backscatter
- Intermediate roughness, can have variations with incidence angle (i.e. water, snow covered ice)



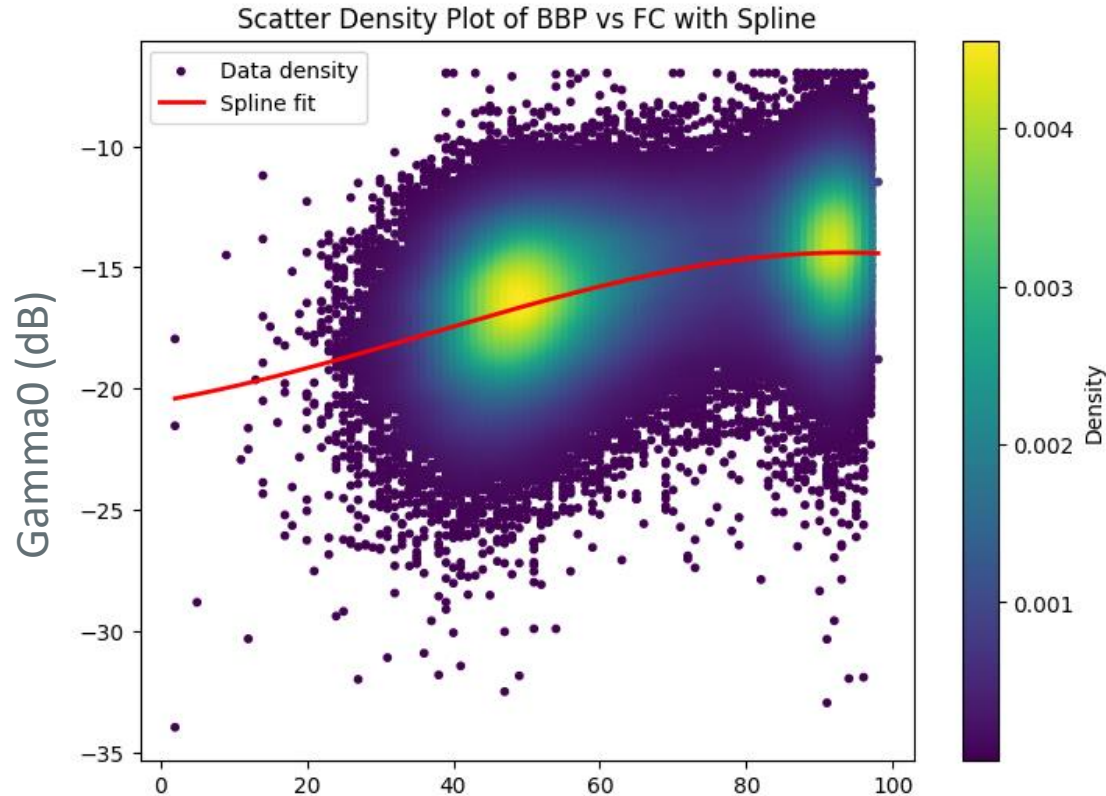
A note on volume scattering

- Signal become saturated with increasing density and/or increasing depth/attenuation

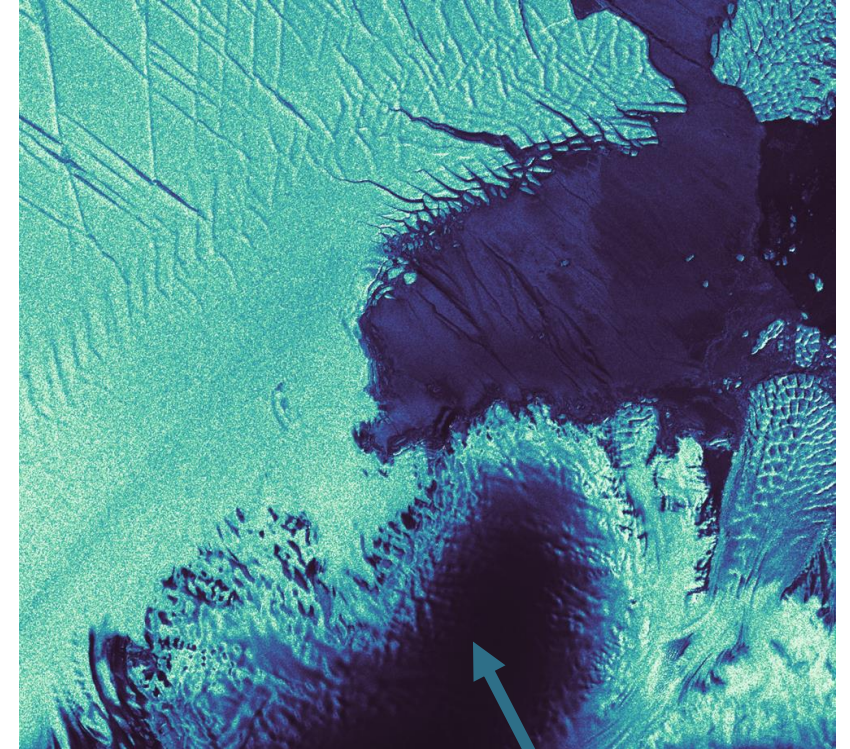
Airborne SAR Injune,
South QLD



Quick plot in Daintree rainforest
DEA LC = woody veg



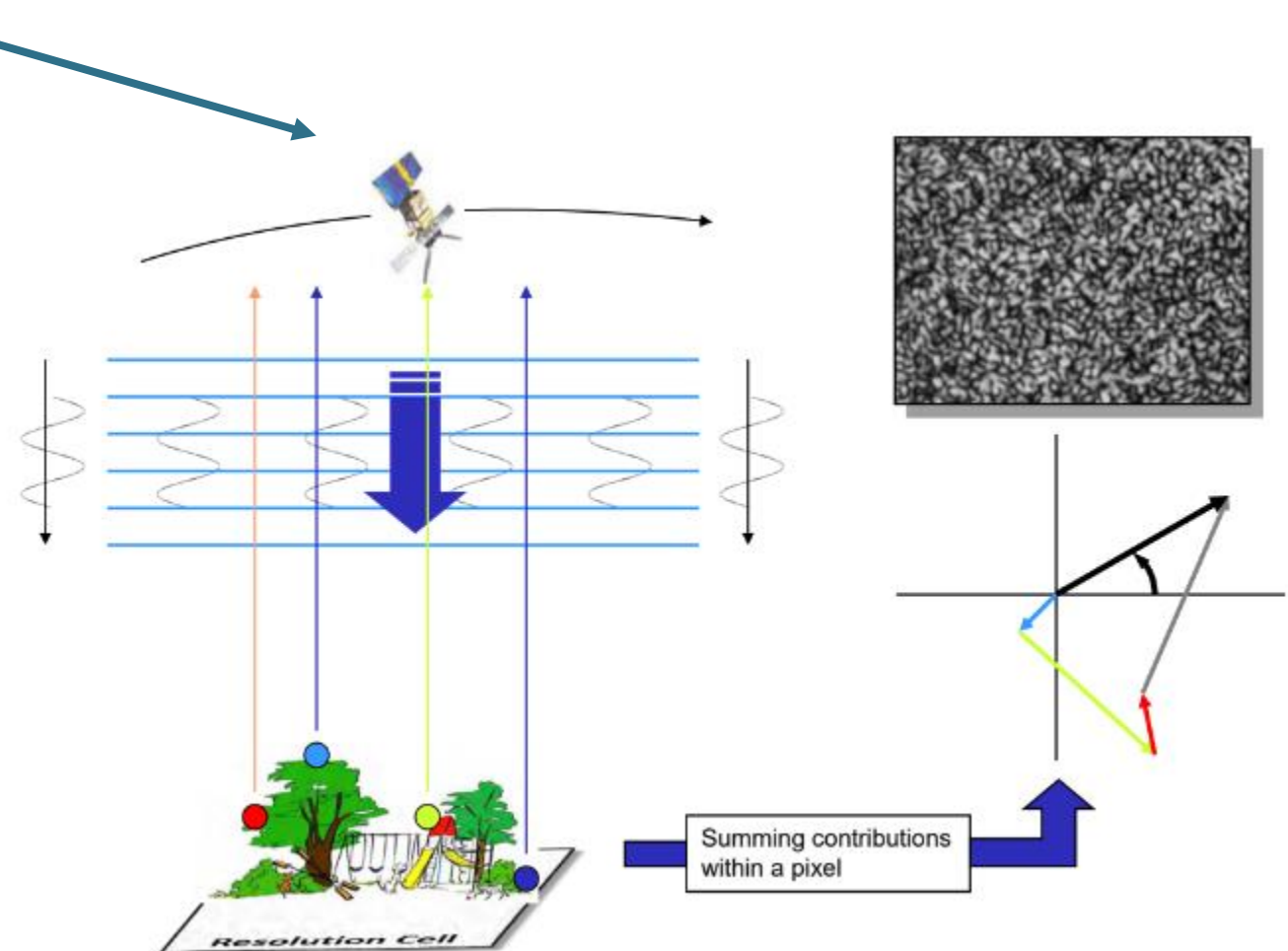
DEA Fractional Cover
What does this relationship look like with
speckle/noise filtering? Is this useful for LC?



Signal almost fully
attenuated by deep snow

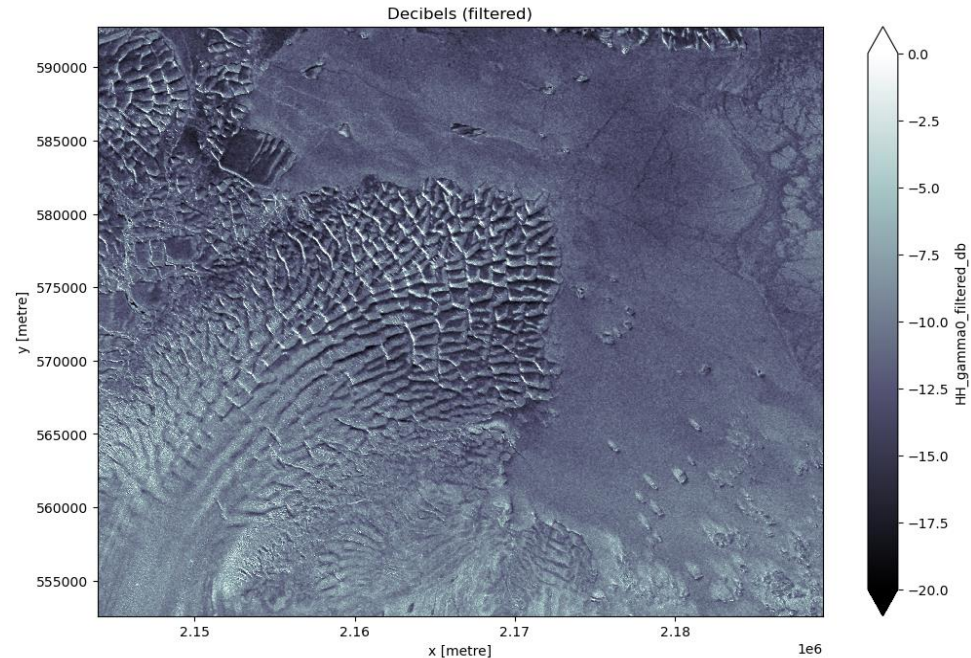
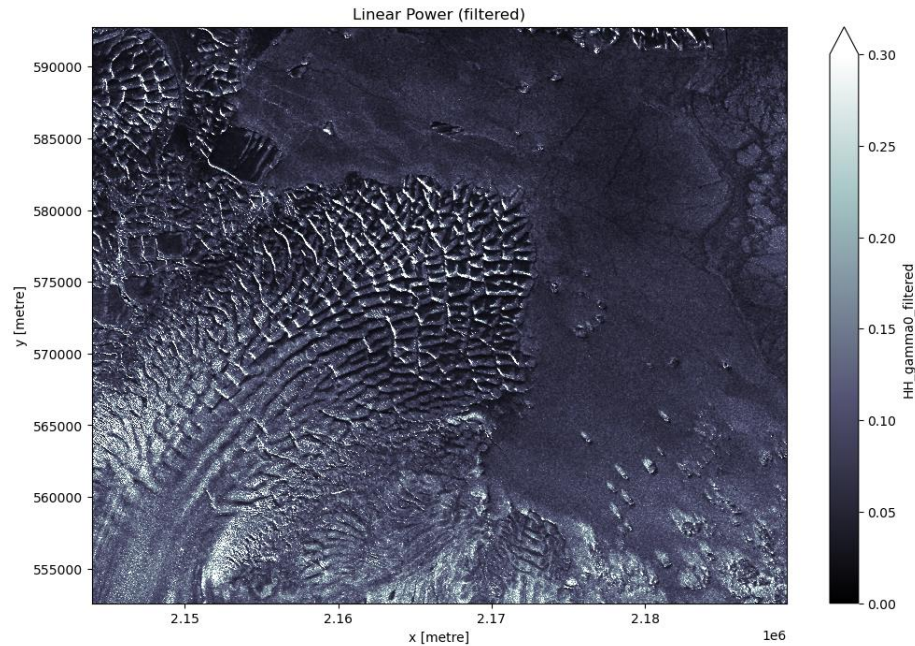
Speckle

- “noisy” appearance of SAR imagery is caused by interference patterns
- Random positive and negative interference of wave contributions from the many individual scatterers within one resolution cell
- Higher backscatter return, higher noise
- Speckle emphasises darker pixels
- Multi-looking/averaging reduces speckle at the expense of resolution
- Other filters:
 - Lee
 - Enhanced Lee
 - Frost
 - Refined Lee
 - Gamma MAP

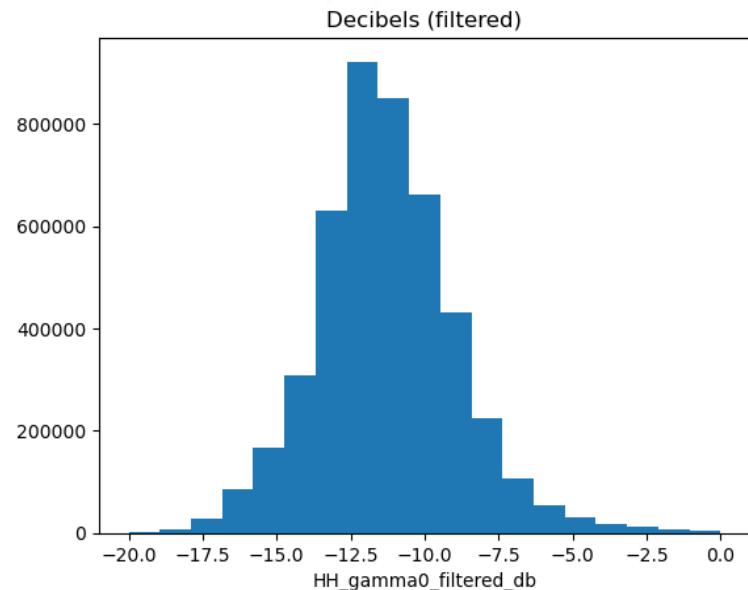
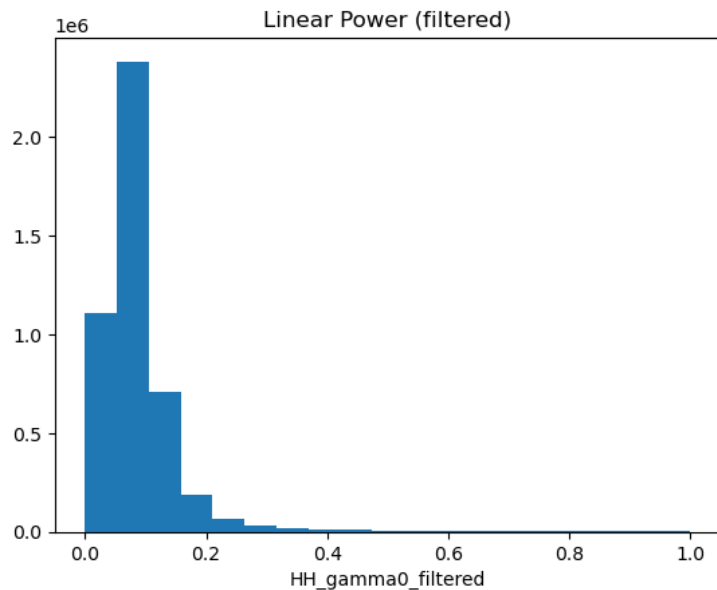


A note on linear and dB scales

$$\text{dB} = 10 \cdot \log_{10}(\text{linear})$$

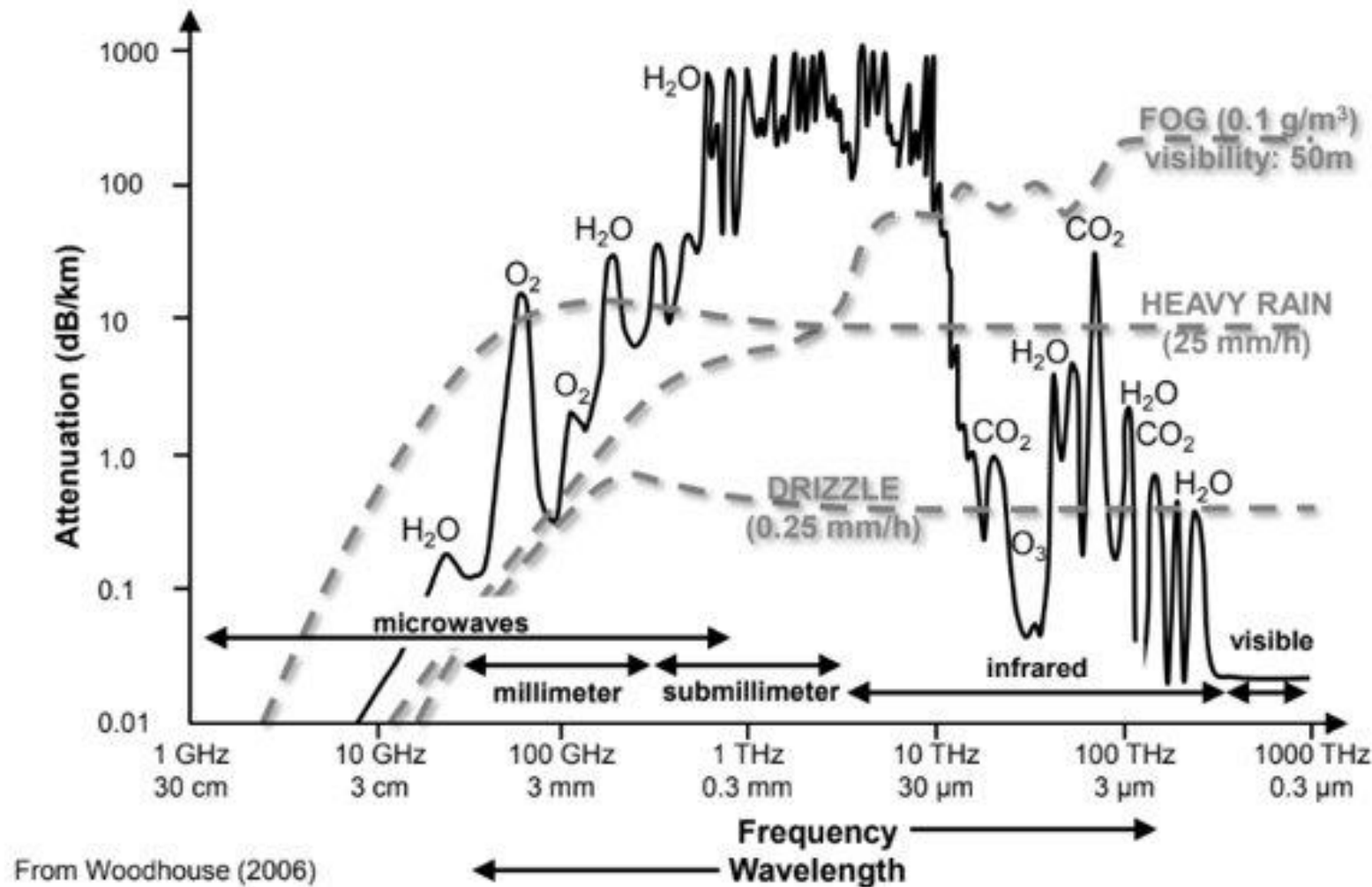


- Trends in dB are not the same as trends in linear and different multi-looking can have a large effect on averages in dB
- Use dB whenever normal distribution assumed – i.e. fitting models/regressions
- Models trained in dB are not necessarily interoperable across different products with different multilooking



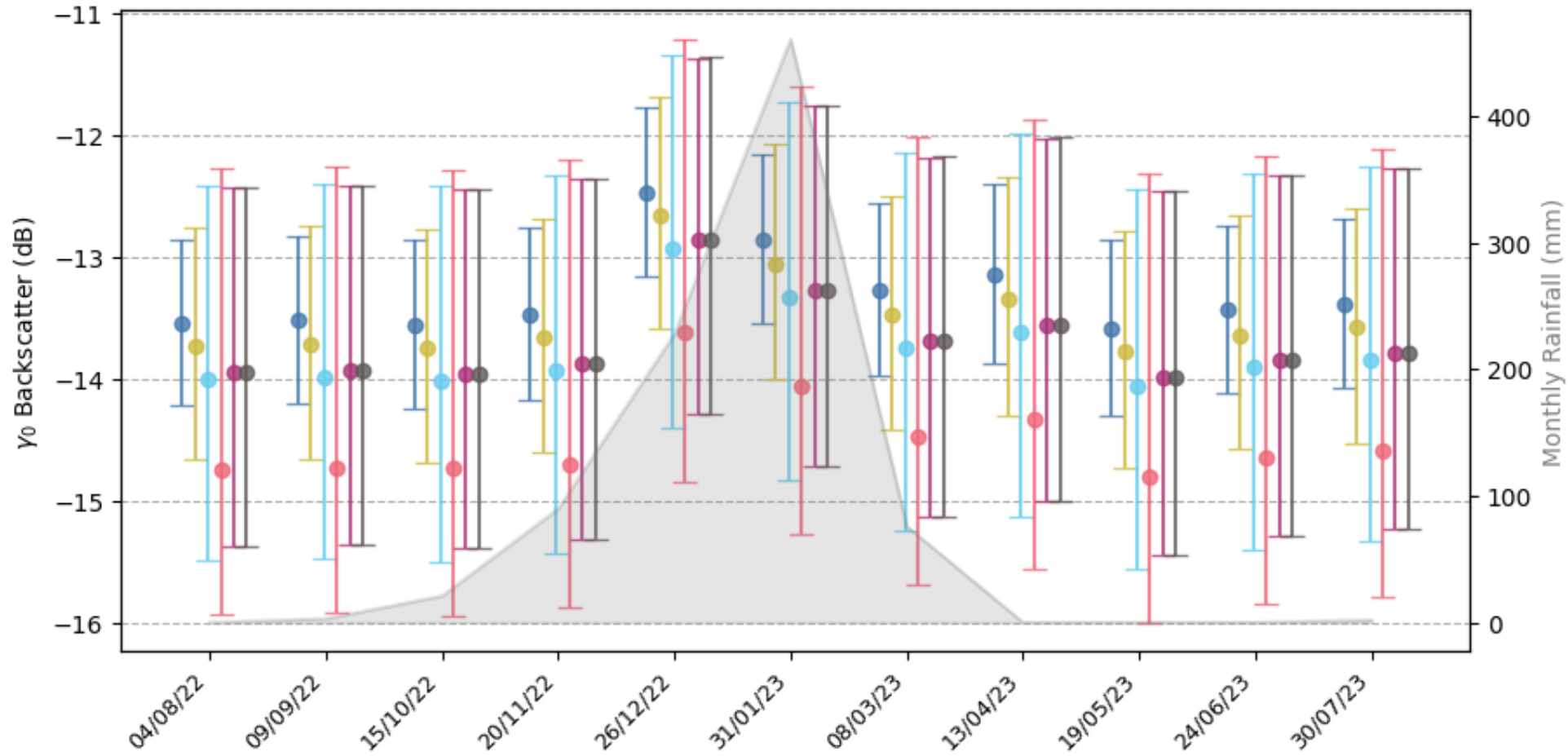
SAR can be impacted by heavy rain

- Water vapour, heavy rain and the ionosphere cause the largest atmospheric interference with SAR and InSAR imagery



SAR can be impacted by heavy rain

- Water on the ground, or in the air?



Variables influencing Backscatter

Sensor:

- Wavelength
- Polarisation
- Look angle (removed/minimised in NRB)
- Resolution (texture)

Target

- Surface roughness and shape
- Local slope and orientation (removed/minimised in NRB)
- Scatterer density within pixel (e.g. biomass)
- 3D distribution (size and shape) of scatterers (i.e. leave size, thickness, grain size)
- Dielectric properties (i.e. moisture, rain)

Producing SAR Analysis Ready Data: Removing Terrain effects

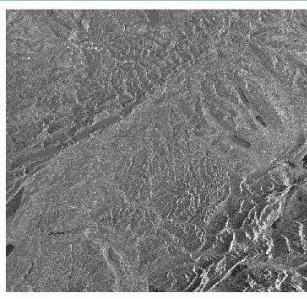
- Normalised radiometrically terrain corrected gamma-nought (γ_0) backscatter (NRB)

Geocoding

Maps from radar to ground coordinates

Radar
coordinates

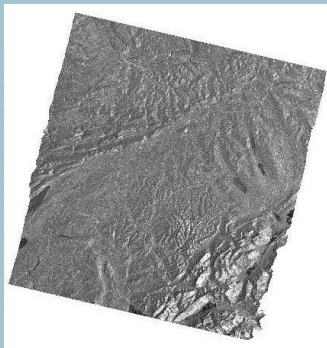
azimuth



range

Ground
coordinates

latitude

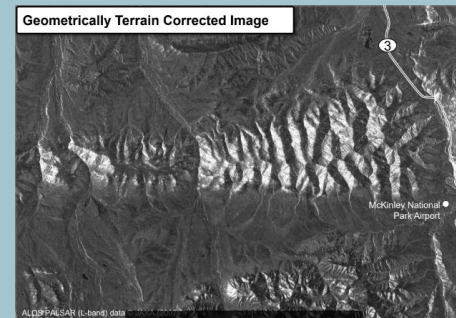
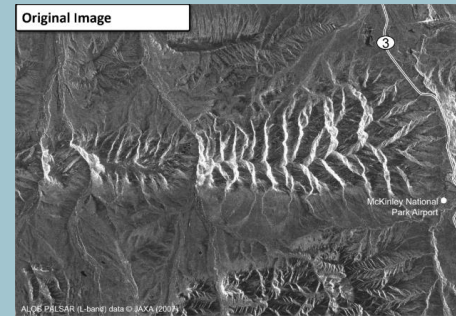


longitude

Geometric Terrain Correction

Corrects foreshortening, layover and shadow

Terrain
distortions

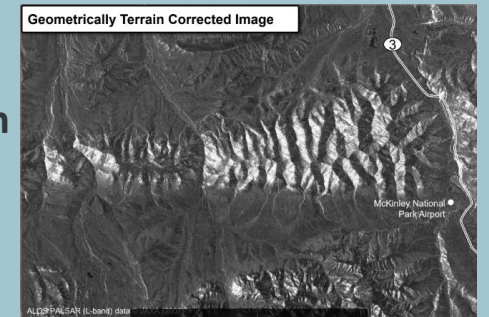


Geocoded
with a DEM

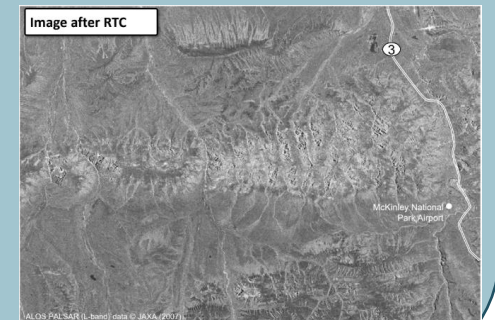
Radiometric Terrain Correction

Uses geocoding to calculate illumination area from the DEM and normalise

Response
changes with
terrain (σ_0)



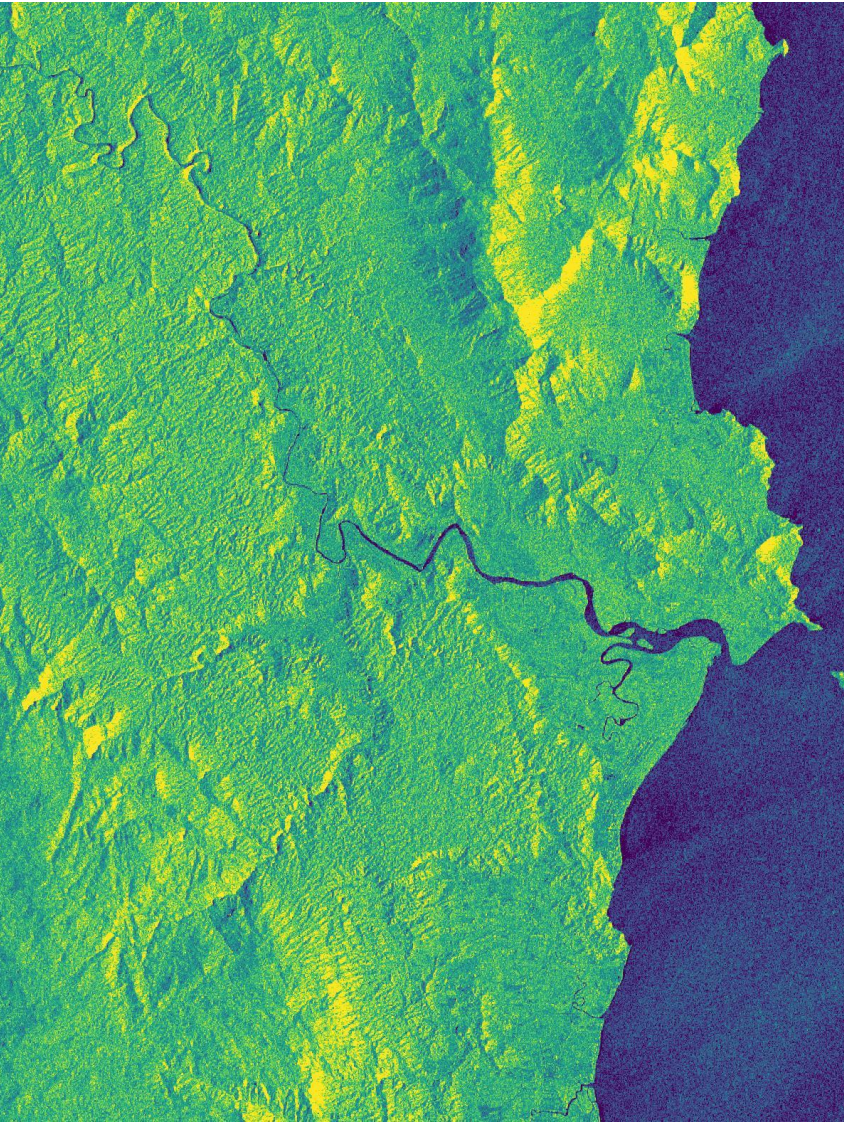
Response
consistent
with terrain
(γ_0)



+ calibration + thermal noise removal + orbital corrections + atmospheric delay corrections + debursting/mosaicking

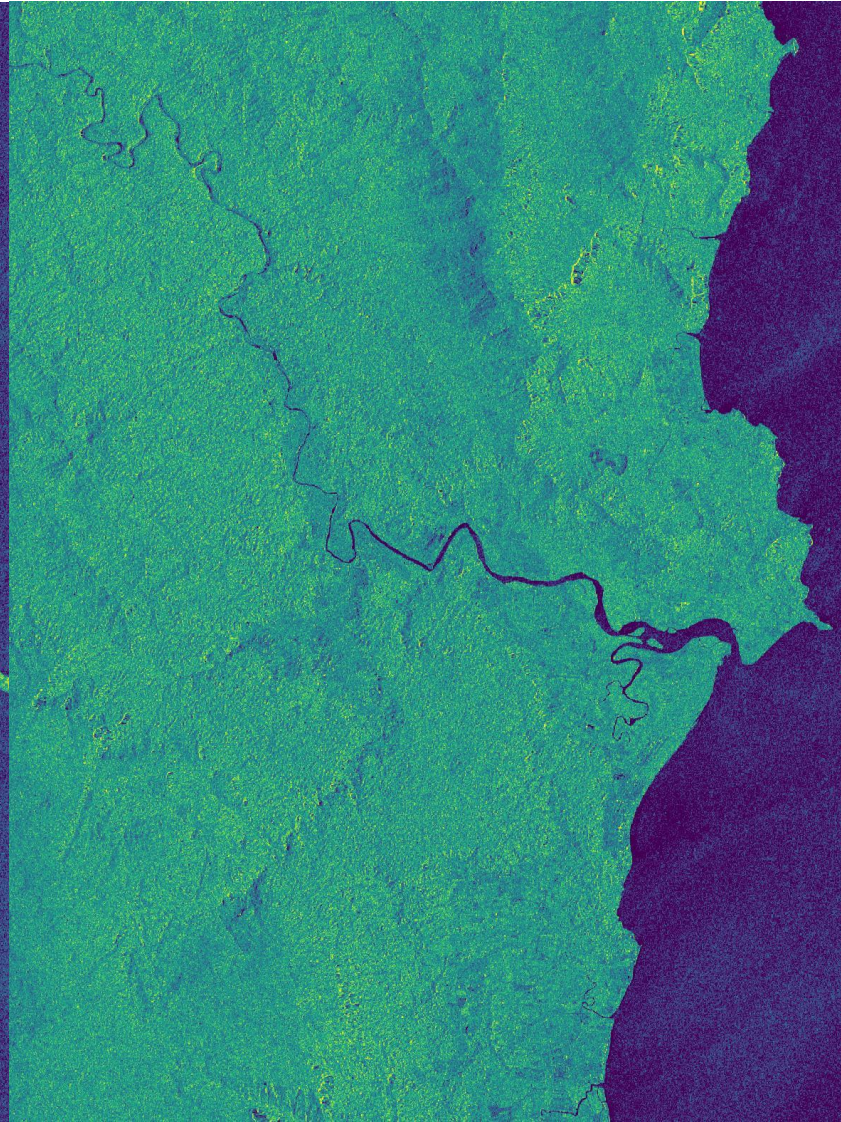
beta0

Backscatter power/intensity



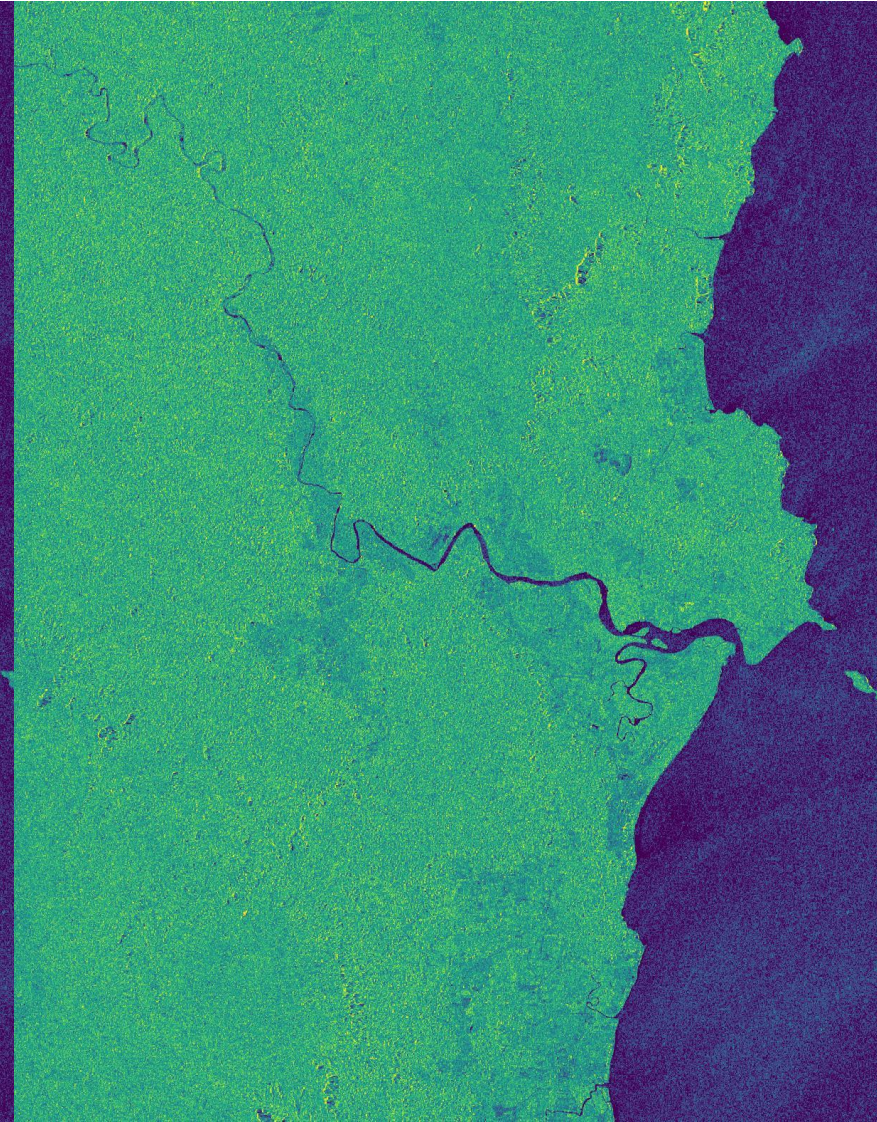
sigma0

Referenced to DEM



gamma0

NRB/Radiometrically Terrain Corrected



Data Conversions and Antarctic DEM Artifacts

ARD

[gamma0]

x

Static layer

[gamma0_to_beta0]

=

ARD

[beta0*]

[gamma0]

x

[gamma0_to_sigma0]

=

[sigma0 (terrain)]

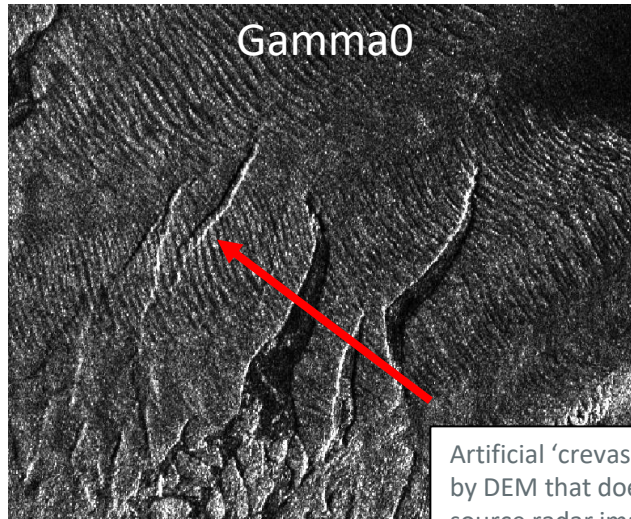
[beta0]

x

sin([incidence_angle])

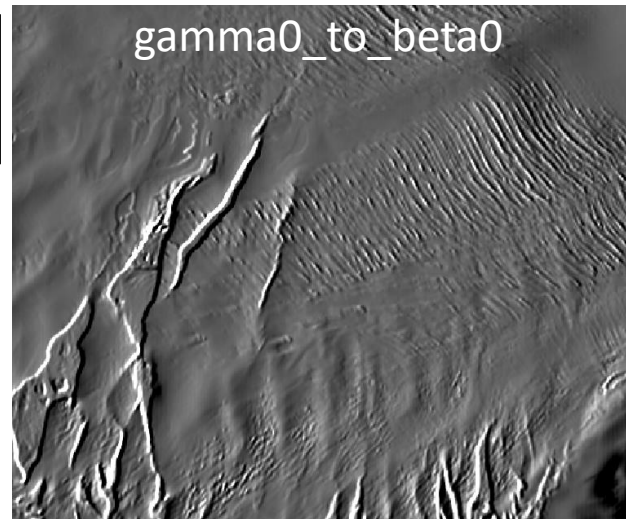
=

[sigma0 (ellipsoid)]

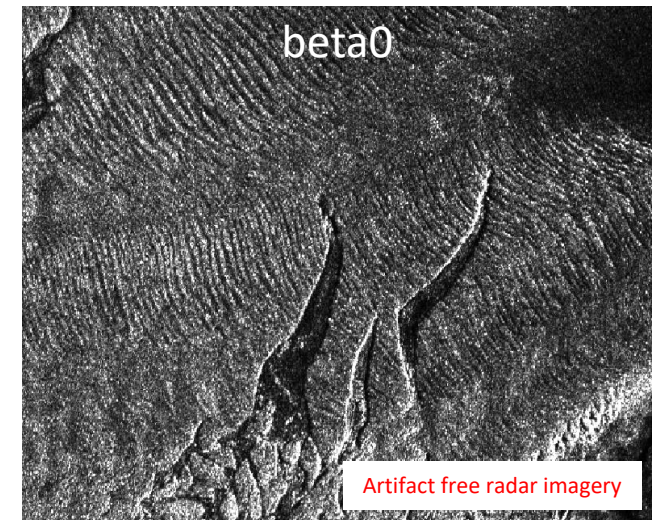


'Undo' radiometric terrain correction from decorrelated DEM

x



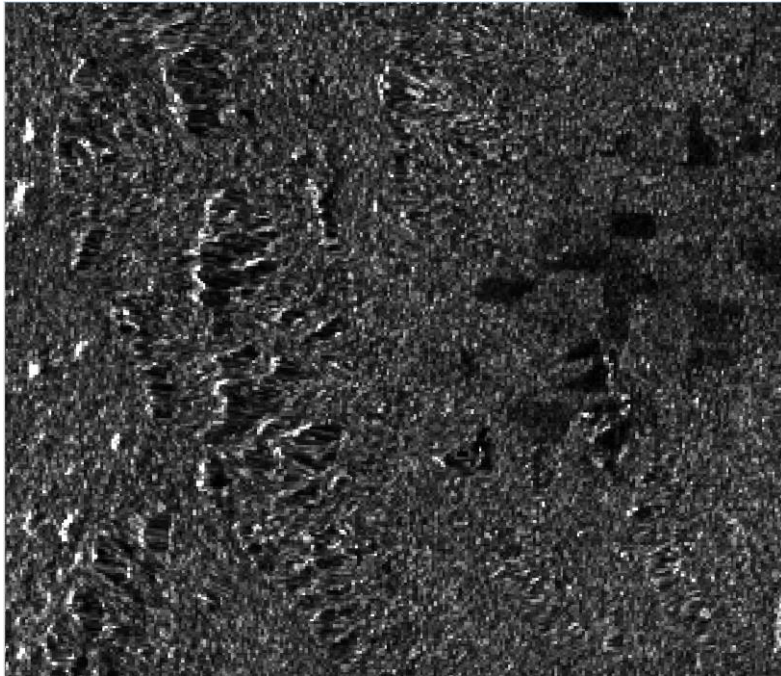
=



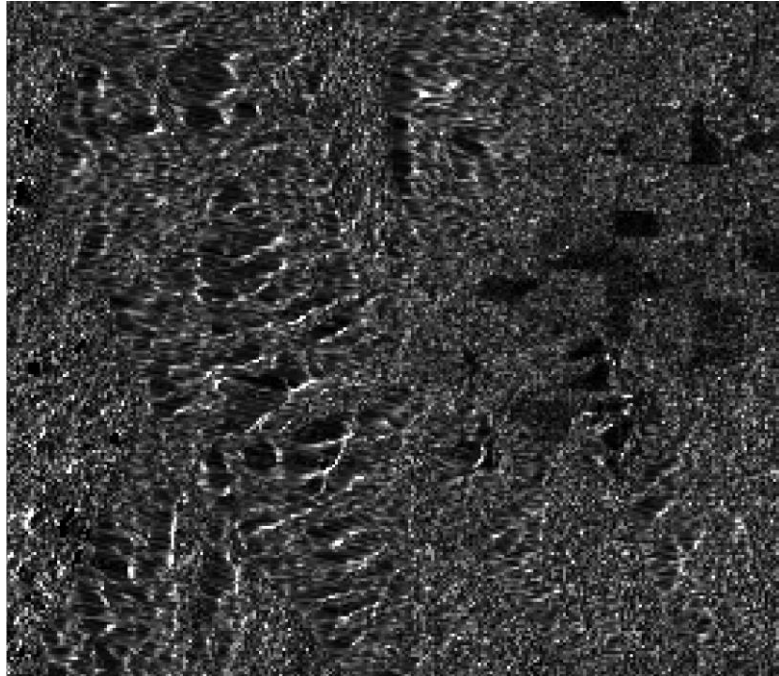
* this is an "approximation" and can have some small discrepancies over terrain due to geocoding/averaging process

Layover/Shadow masks

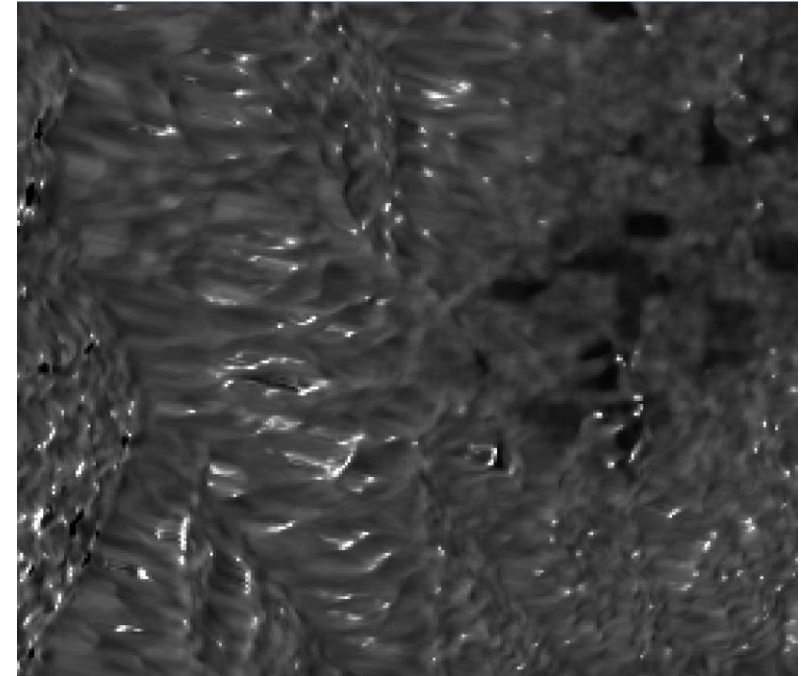
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CSIRO unfiltered

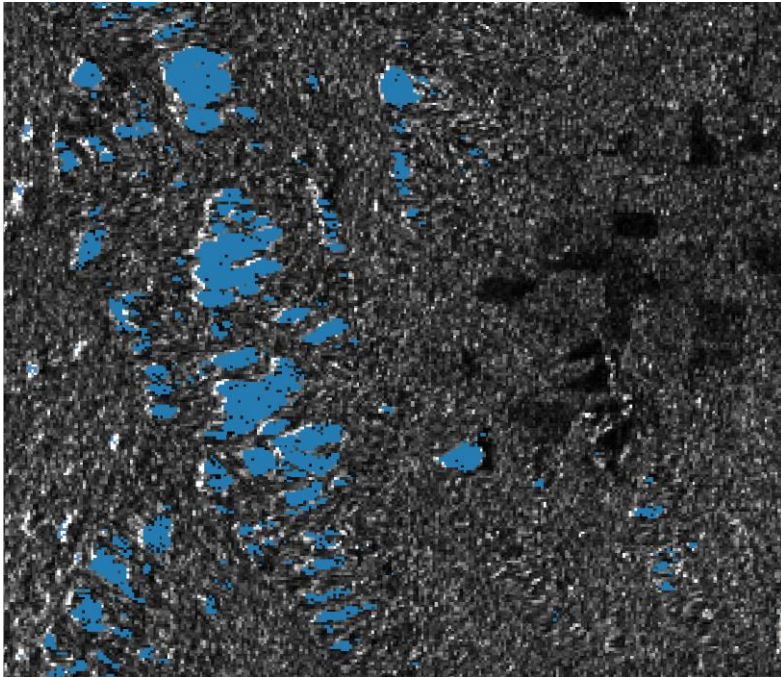


CSIRO filtered

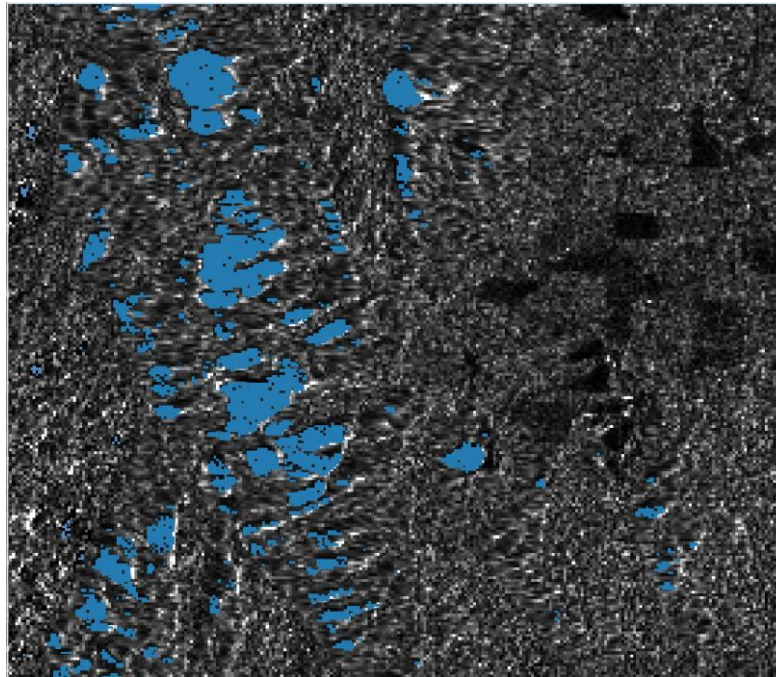


Layover/Shadow masks

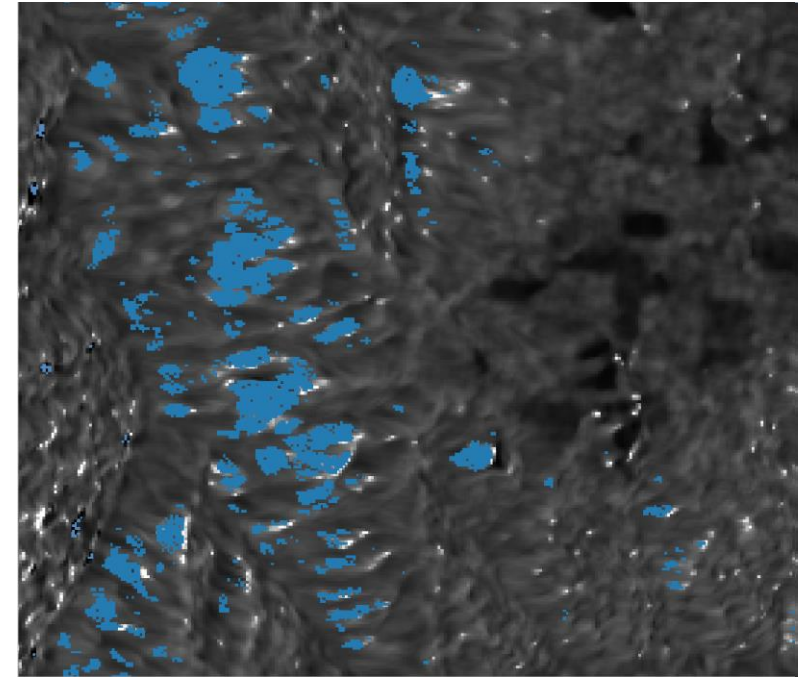
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CSIRO unfiltered



CSIRO filtered



Topographic Quality

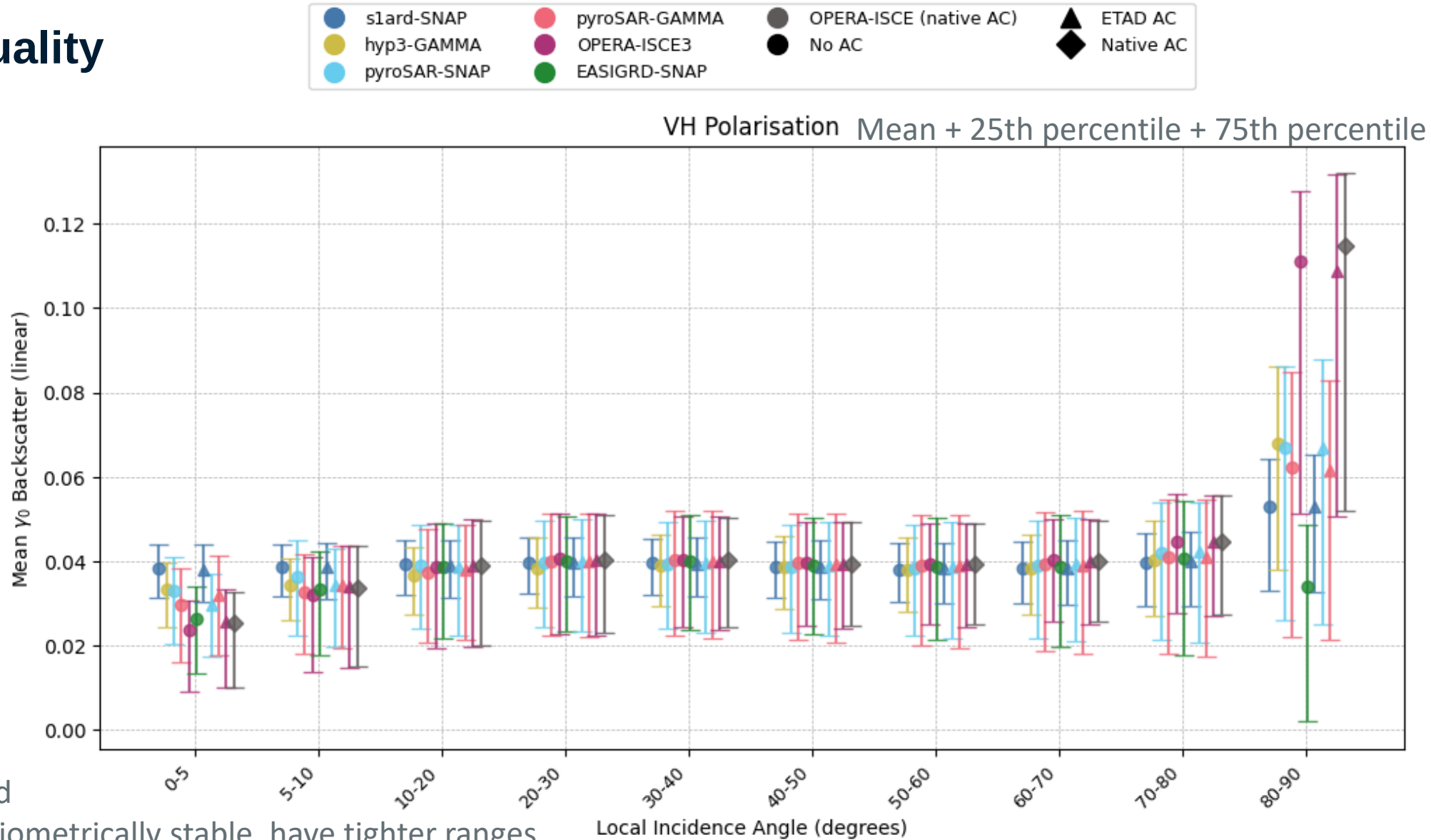
Local Incidence Angle

There is no trend between observations and local incidence angle over rugged terrain with invariant land cover

GTC + RTC

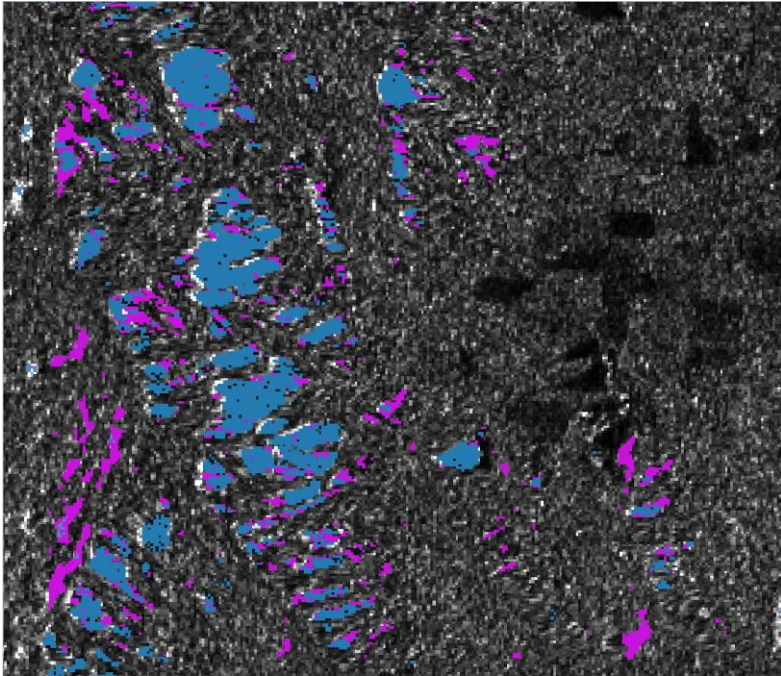
Results

- 20° – 70° : similar trend
- OPERA-ISCE3 most radiometrically stable, have tighter ranges
- Differences do from < 0.1 db to 0.5 db for higher LIA (20-70°)
- Masking additionally for LIA < 10° and > 70/80° seems effective

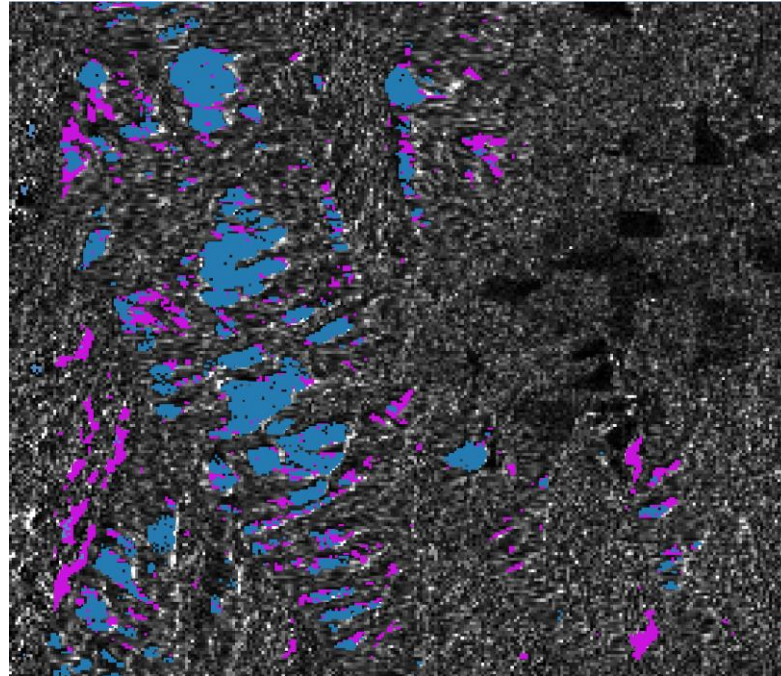


Layover/Shadow masks

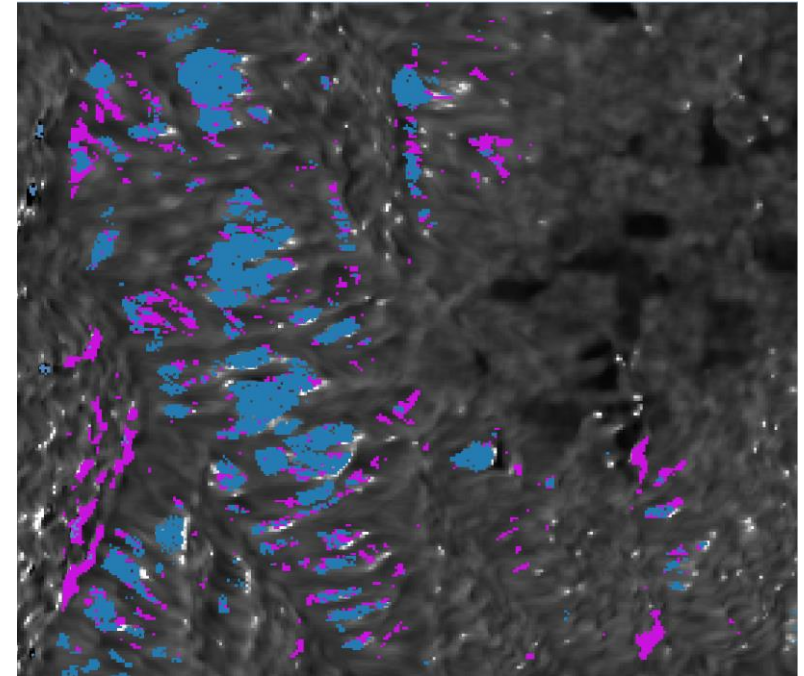
Geoscience Australia



CSIRO unfiltered



CSIRO filtered



Synthetic Aperture Radar (SAR) capability adds value



- Observation at night
- Increased observation under cloud cover
- Added information



- Observation in polar winter
- Increased observation under cloud cover
- Added information

Application	Added value	Maturity
Floods/Waterbodies	Increased observations	Advanced
Soil/Ground Moisture	Unique to SAR	Advanced
Coastal erosion/shoreline	Increased observations. Map coastal inundation.	Developed
Land cover and vegetation	Distinguish woody from non-woody classes	Developed
Urban Cover	Distinguish urban from bare-land	Developed
Coastal habitats	Distinguish saltmarsh, mangrove, supratidal	Experimental
Historical burn area	Detect under canopies	Experimental
Crop classification	Distinguish crops	Experimental

Application	Added value	Maturity
Ice velocity and basal melt	Increased observations	Advanced
Ice shelf and calving	Increased observations	Advanced
Iceberg tracking	Increased observations	Advanced
Crevasse detection	Under-snow detection	Advanced
Snow classification and melting	Unique to SAR	Advanced
Sea-ice and fast-ice	Increased observations	Developed
Sub-glacial lakes	Increased observations and freeze state	Developed

Common SAR Bands/Frequencies

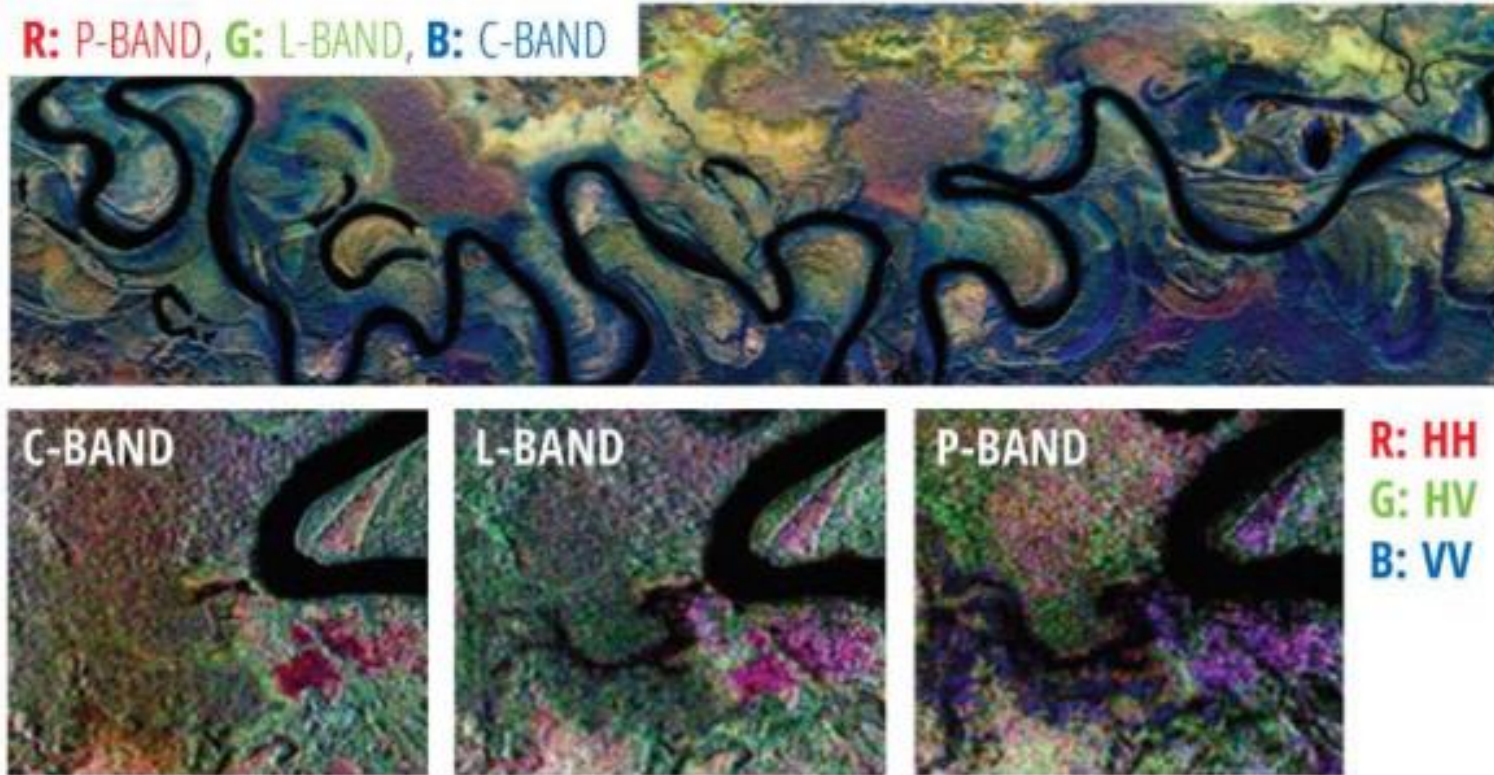
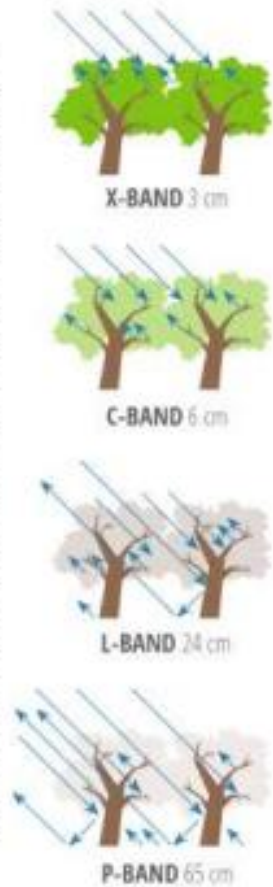
Band	Frequency f_0		Wavelength $\lambda = c/f_0$		Typical Application
Ka	27 – 40	GHz	1.1 – 0.8	cm	Rarely used for SAR (airport surveillance)
K	18 – 27	GHz	1.7 – 1.1	cm	Rarely used for SAR (H ₂ O absorption)
Ku	12 – 18	GHz	2.4 – 1.7	cm	Rarely used for SAR (satellite altimetry)
X	8 – 12	GHz	3.8 – 2.4	cm	High-resolution SAR (urban monitoring; ice and snow; little penetration into vegetation cover; fast coherence decay in vegetated areas)
C	4 – 8	GHz	7.5 – 3.8	cm	SAR workhorse (global mapping; change detection; monitoring areas with low to moderate vegetation; improved penetration; higher coherence)
S	2 – 4	GHz	15 – 7.5	cm	Little but increasing use for SAR-based Earth obs. ; agriculture monitoring (NISAR will carry S-band; expands C-band applications to higher vegetation density)
L	1 – 2	GHz	30 – 15	cm	Medium resolution SAR (Geophysical monitoring; biomass and vegetation mapping; high penetration; InSAR)
P	0.3 – 1	GHz	100 – 30	cm	Biomass estimation. First P-band spaceborne SAR will be launched ~2020; vegetation mapping and assessment. Experimental SAR.

Remember, wavelength penetration can cause differences in polarimetry

SAR Signatures of Vegetation at Various Frequencies and Polarizations



Increase of penetration depth with wavelength



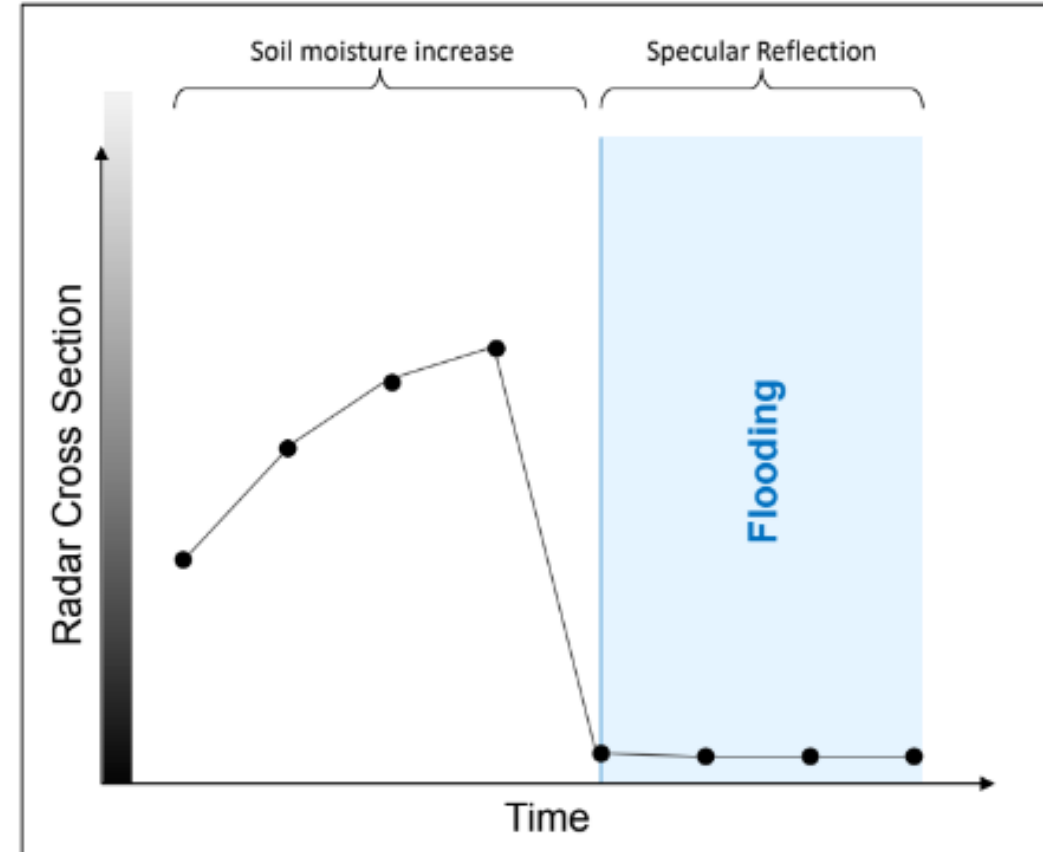
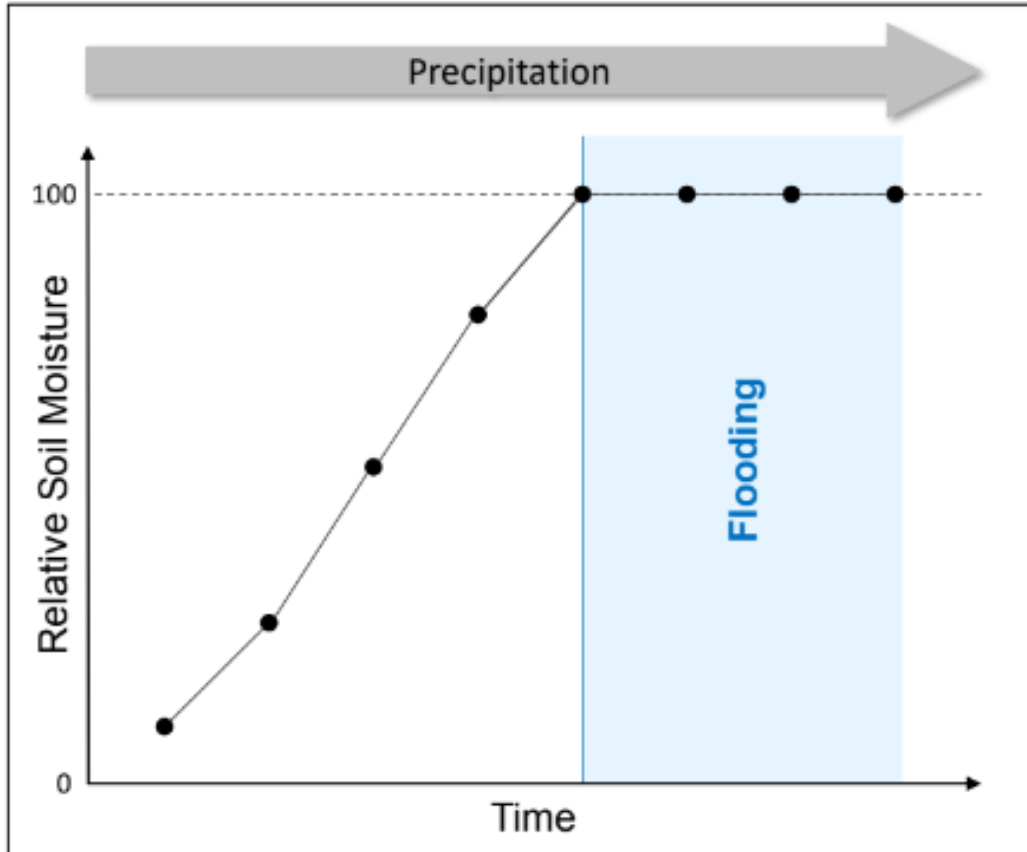
Polarimetric signature change and sub-canopy features are exposed as wavelength increases

SAR Handbook: Comprehensive Methodologies for Forest Monitoring and Biomass Estimation. Eds. Flores, A., Herndon, K., Thapa, R., Cherrington, E. NASA. 2019.
DOI: 10.25966/hbm1-ej07

Surface Water Mapping

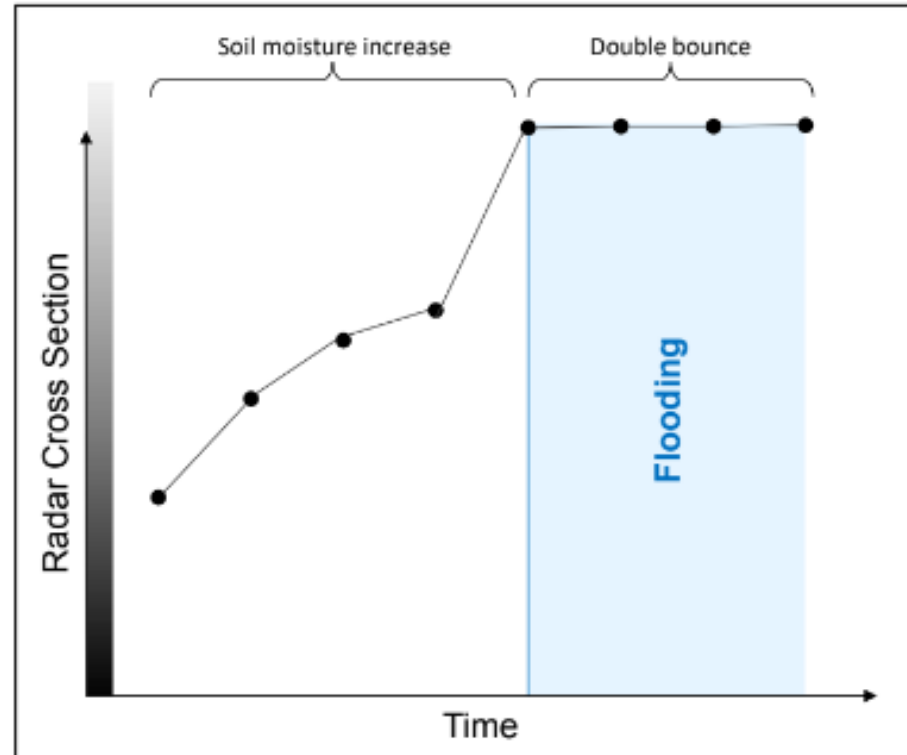
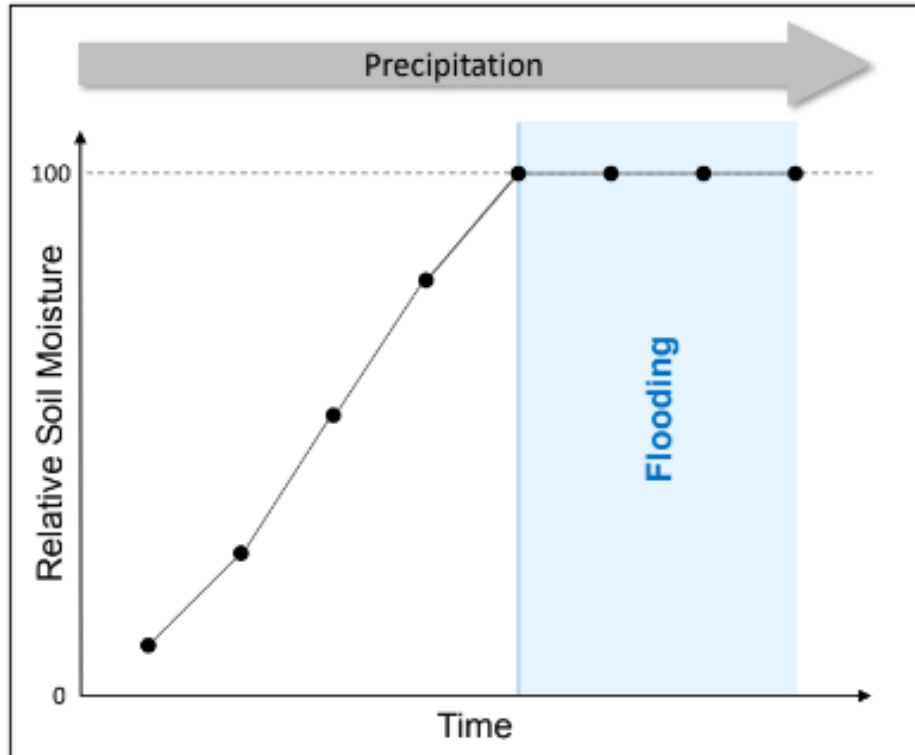
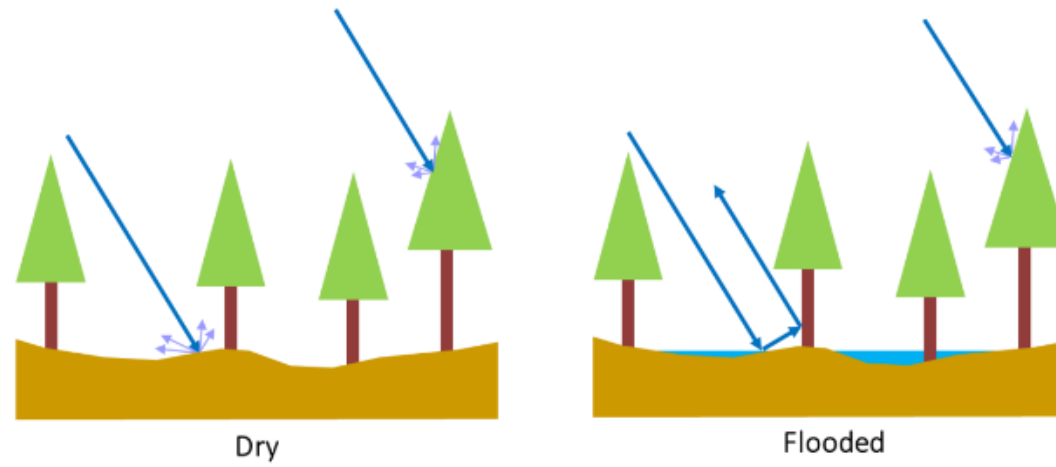
- Refresh: Surface water is dark, due to specular scattering
- Increases in soil moisture increases backscatter

Open lands:



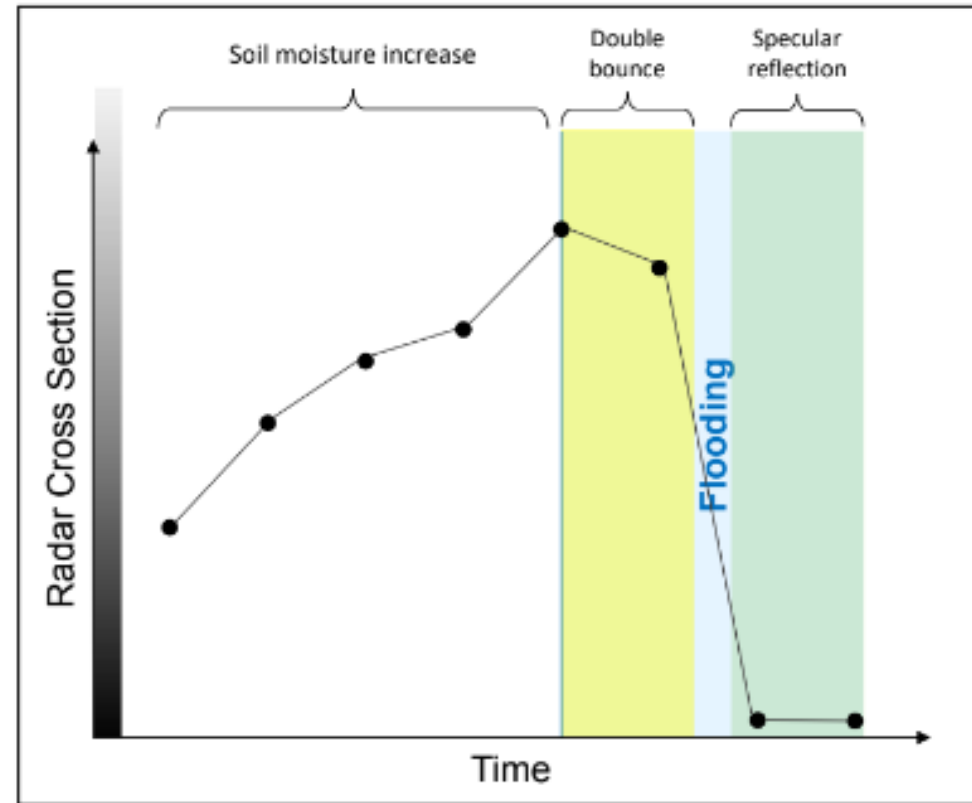
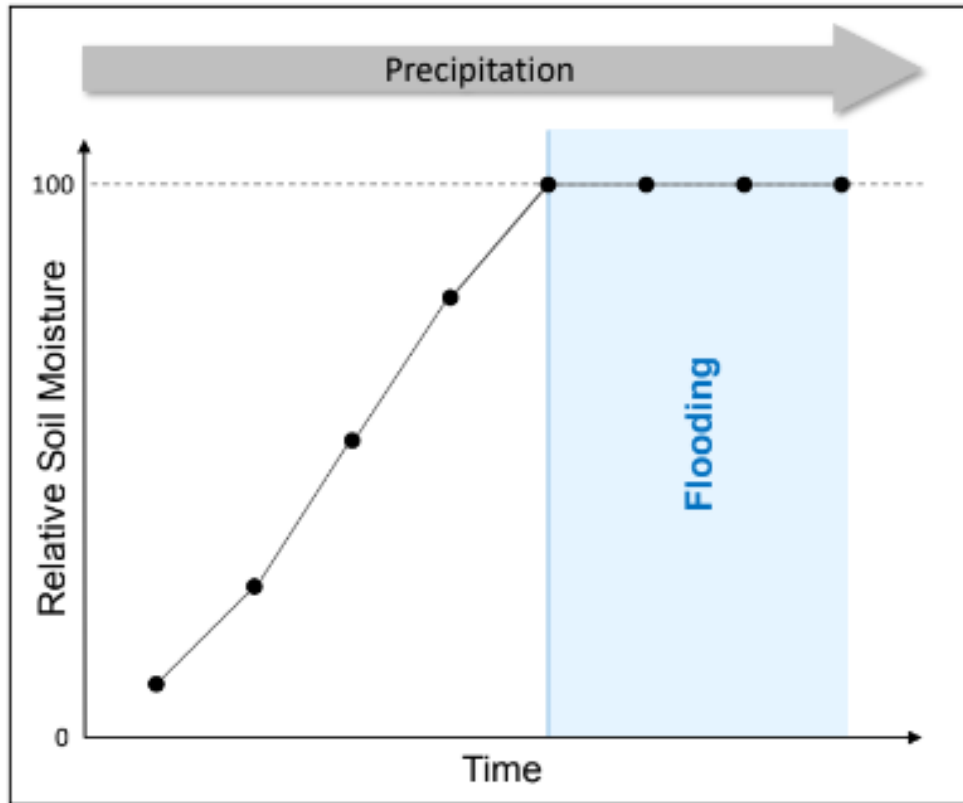
Surface Water Mapping

Vegetated Canopies



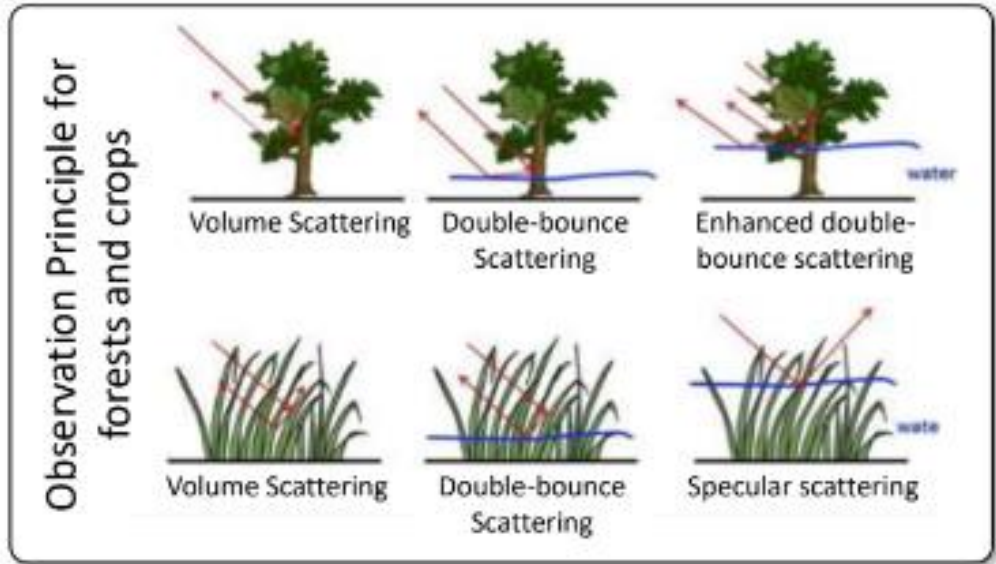
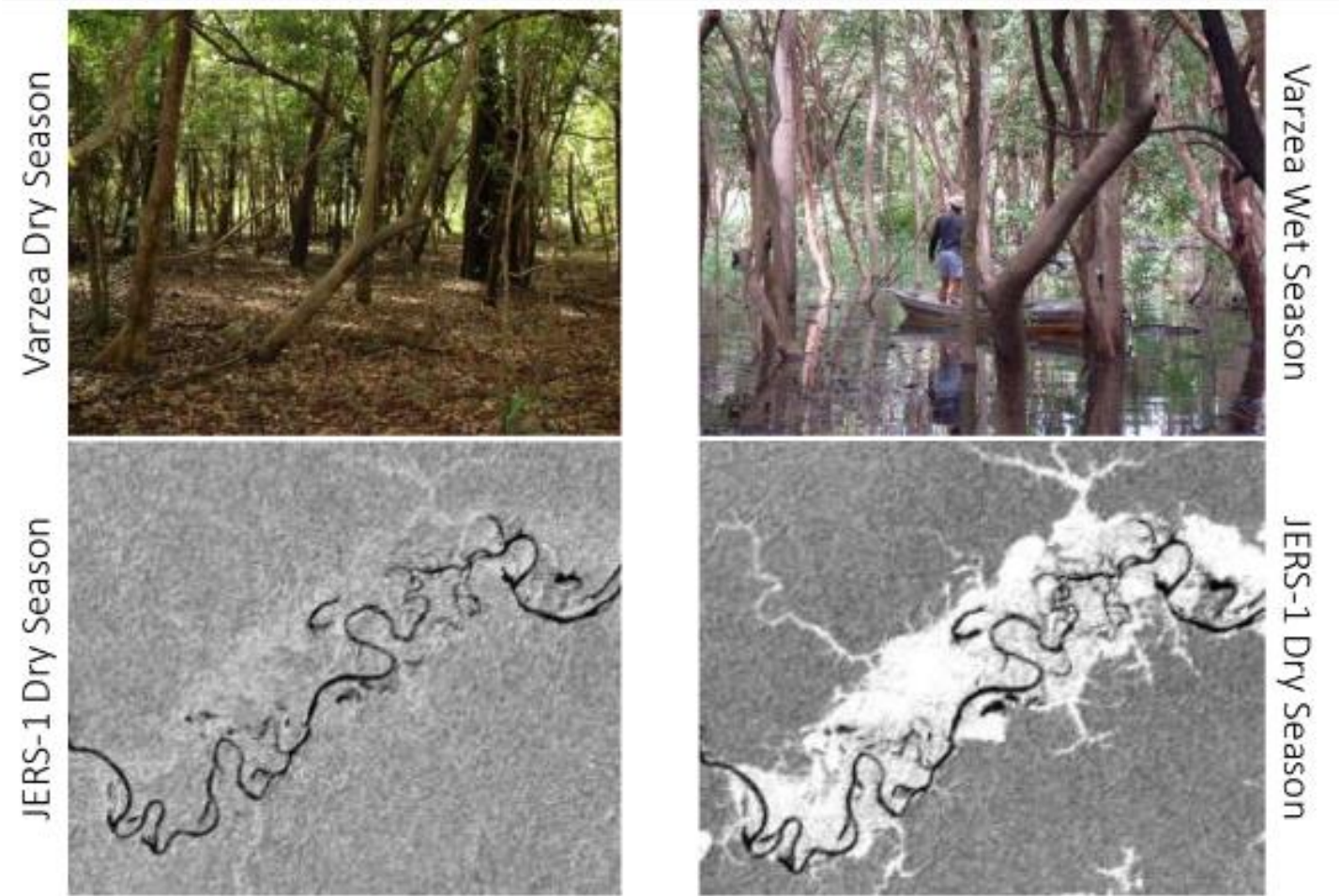
Surface Water Mapping

Cropland (ie. short trees that become covered)



Surface Water Mapping

- L-band is best for inundation mapping

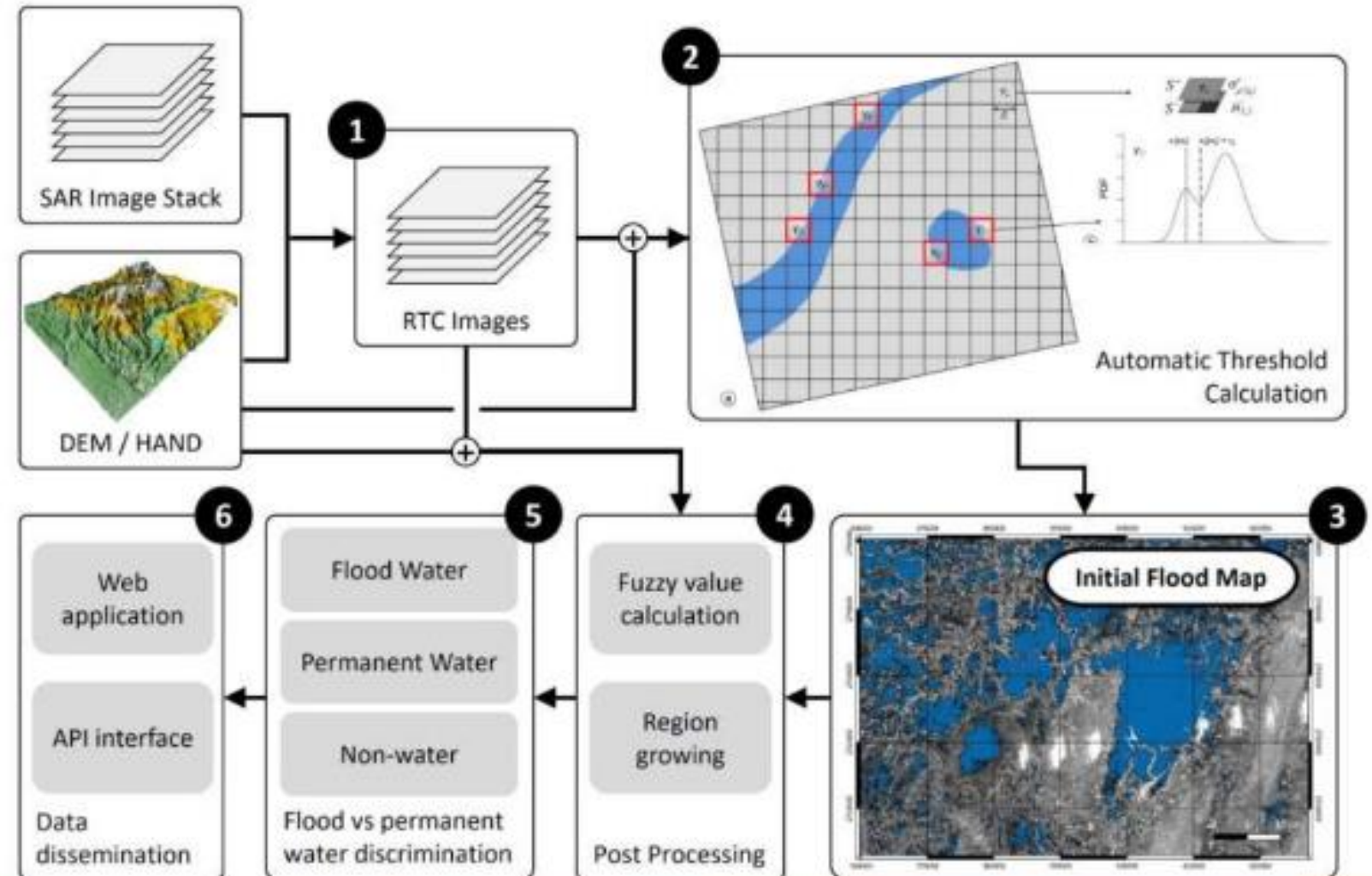


Surface Water Mapping: HydroSAR

- Threshold-based method, also applies logic to remove spurious areas

HydroSAR water mapping approach composed of 6 steps:

1. Image Geocoding and Calibration (RTC Processing)
2. Automatic and adaptive threshold calculation
3. Initial flood map creating
4. Post-processing to remove false alarms
5. Discrimination of permanent and flood-related water
6. Data dissemination



Surface Water Mapping: Hydrosar

- Flood water depth available using a “bathtub model”, and height above nearest drainage (HAND) calculation

